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Technical Report No. 85

Enhanced Test Methods for Visible Particle Detection and Enumeration on Elastomeric Closures and Glass Containers

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1.0 Introduction

Visual inspection of filled injectable products is an important part of the control of the manufacturing process. It offers a means to improve quality by removing noncompliant units and provides information to drive continuous process improvement.

Visual Inspection (VI) may be accomplished in several ways: manually (MVI) with the unaided eye under controlled conditions of lighting and time; semi-automated visual inspection (SAVI), which combines 1) automated handling of the containers to be inspected with human vision, and 2) a means to determine whether an individual unit should be accepted or rejected; and thirdly, automated visual inspection (AVI), which offers a validatable method using digital cameras to acquire images of the product and computational tools to analyze those images against preset criteria to determine if the unit is acceptable. While MVI is the reference method described in all the pharmacopoeias, all three categories are acceptable with appropriate qualification or validation.

For MVI, the visible range for particles in many filled injectable products is generally agreed to begin at 100 µm based on studies in clear glass vials ***(1).*** However, not all particles of this size can be reliably detected. Nominal probability of detection (PoD) for 100 µm particles is often 40-50% with PoD increasing with increasing particle size. Single particles greater than 200 µm in clear solution and in 10 mL clear vials typically achieve a PoD >95% ***(2).***

To reliably meet visible particle acceptance criterion for filled products, careful control of the components used in the process is required. Criteria must generally be tighter than that of the finished product to compensate for the cumulative effects of multiple components (e.g., container and elastomeric seal), potential sources from the formulation and filling equipment, and the manufacturing environment. Today, most components are not subjected to tighter requirements as part of their incoming inspection and acceptance for use. However, specifications that meet stricter requirements are most critical for ready-to-sterilize (RTS) and ready-to-use (RTU) components since they do not undergo further particle removal steps.

1.1 Purpose

This technical report addresses test and inspection methods for visible particles in primary packaging components for injectable products specifically glass vials and syringes, and the elastomeric closures used to seal them. This document is most relevant for RTS and RTU components since these are not subjected to a further cleaning process immediately before use. These studies and methods were developed as part of a larger initiative to eliminate visible particles in injectable products. An overview of this initiative was published ***(3)*** previously, and a detailed analysis of particle entry points to the manufacturing process was subsequently published ***(4).***

1.2 Scope

The use of particle, particulate, and particulate matter all have the same meaning when used within this report. They are used in various global standards and compendia with equivalent meaning. In general, this term applies to foreign material which is unintended and either removable from the surface of a primary packaging component or freely moving within a filled and sealed pharmaceutical dosage form.

The test and inspection methods discussed within this report are the first step in a standard setting process. Once test and inspection methods are available, accepted, and implemented, they can be used to assess process capability and establish meaningful process control limits. Without reliable measurements, reliable control limits are not possible.

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2.0 Glossary and Abbreviations

Definitions have been provided to help clarify the concepts discussed in this technical report. While some of these definitions vary among companies, the definitions described below are consistent for use within this document.

Area
Measured surface area of a flat projection of a particle.

Aspect Ratio
The length of a particle or fiber divided by its width.

Extrinsic Particle or Fiber
Foreign material that comes from outside the primary process. Often these are from the manufacturing environment or incomplete cleaning of components. They are uncontrolled.

Feret Min and Max
Feret Min is the minimum distance between parallel tangents at opposing particle borders. Feret Max is the maximum distance between parallel tangents at opposing particle borders.

Intrinsic Particle
A particle that comes from within the primary process. These are qualified product contact materials and are often associated with the primary packaging components. They are unplanned but not unexpected.

Length
The measure of the distance along its longest axis.

Light obscuration (LO)
Analytical technique measuring the size and quantity of subvisible particles.

Membrane/microscopy (MM)
Analytical technique to collect particles from a liquid sample on a membrane filter followed by manual examination (sizing and counting) with a microscope.

Particle
Foreign material which is unintended and either removable from the surface of a primary packaging component or freely moving within a filled and sealed pharmaceutical dosage form. The term particle is used interchangeably with "particulate" and "particulate matter" without any intended difference in meaning.

Particulate
See Particle.

Perimeter
Measured length of the outer edges of a particle.

Probability of detection (PoD)
The likelihood that a given particle will be seen and removed with a specific inspection method and under specific conditions. Usually expressed as a percentage (0-100%) but may also be expressed as a probability (0-1).

Rect A
The longest side length when a rectangle is constructed around the particle.

Ready-to-sterilize (RTS)
Washed components supplied in a package suitable for sterilization prior to use with sterile injectable products.

Ready-to-use (RTU)
Washed and sterilized components supplied in a package suitable for transfer into an aseptic processing area and used with sterile injectable products without further processing.

Shape factor (SF)/Circularity
Provides information on the roundness of a particle. The SF value is between 0 and 1, where 1, represents a circle. A is the area of the particle and p is the perimeter. $SF = \frac{4\pi A}{p^2}$

Subvisible particle
A particle with a size <100 µm in diameter.

Visible fiber
A fiber-like particle with an aspect ratio ≥5 and a length >300 µm.

Visible particle
A particle with an aspect ratio of <5 and a length >100 µm.

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2.1 Abbreviations

AQL	Acceptable Quality Limit
AVI	Automated Visual Inspection
cGMP	Current Good Manufacturing Practices
DOE	Design Of Experiments
EP	European Pharmacopeia or Pharm Europa
FDA	United States Food and Drug Administration
IM	Intramuscular
ISO	International Organization for Standardization
IV	Intravenous
JP	Japanese Pharmacopeia
LED	Light Emitting Diode
LO	Light Obscuration
min	Minutes
MM	Membrane/Microscopy
MVI	Manual Visual Inspection
NMT	Not More Than
PMF	Pharmaceutical Manufacturers Forum
PoD	Probability Of Detection
RPM	Revolutions Per Minute
RTS	Ready-To-Sterilize
RTU	Ready-To-Use
SAVI	Semi-Automated Visual Inspection
sec	Seconds
SF	Shape Factor
TR	Technical Report
USP	United States Pharmacopeia
VI	Visual Inspection

3.0 Current State of Visual Inspection for Particles

A review of recalls of injectable products over the last ten years provides a useful measure of the current state of manufacturing and particle control. **Figure 3.0-1** shows this information for product regulated by the U.S. Food and Drug Administration (FDA). The major contributors to these recalls are lack of sterility/sterility assurance and particles.

Particle recalls alone in **Figure 3.0-2** shows a notable upward trend in 2013 with a peak in 2014. This was driven by an increased occurrence of both glass and metal particles. This resulted in greater awareness by both manufacturers and regulators and contributed to the higher number of particle recalls in following years. This situation was exacerbated by the nonspecific requirements for particles found in the global pharmacopeias but improved with the official publication of USP <790> in 2014 ***(6).*** At present, annual recalls for particles have returned to historical levels, but still account for a significant amount of product removed from the market and does not consider lots rejected while still within the control of the manufacturer.

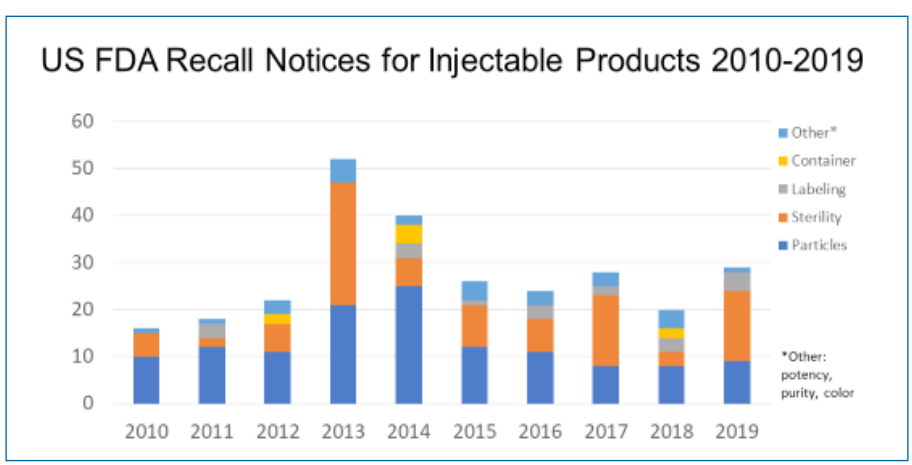

Figure 3.0-1 FDA Recall Notices For Injectable Products, Years 2010-2019 ***(5)***

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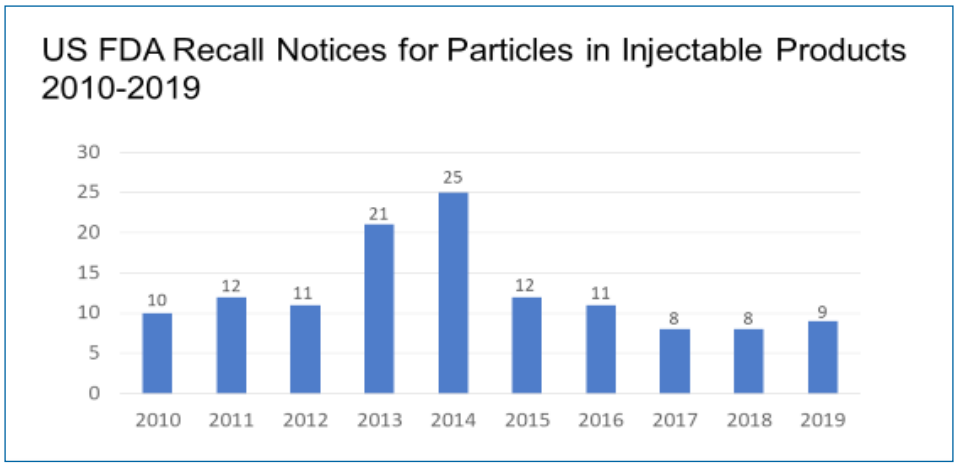

Figure 3.0-2 FDA Recall Notices For Particles In Injectable Products, Years 2010-2019 ***(5)***

3.1 Patient Risk Assessment

All parenteral drug preparations contain particulate matter. Particles ultimately administered to a patient may originate from the primary packaging components or may enter through the manufacturing process and during administration from the syringe or infusion set apparatus. Therefore, the potential patient particulate exposure burden derives from the sum of these sources.

The clinical implications of visible particulate matter in injectables have been reviewed in several recent publications ***(7,8).*** Historically the presence of unintended visible particulate matter has been regarded as a quality defect which can impact safety resulting in additional batch-related actions, including batch reinspection, rejection, or recall. Visible product defects such as cracked or leaking containers or product discoloration often point to a product quality and possible safety issue. While the presence of a rare or isolated visible particle induces a similar instinctive response, the potential safety implication requires a comprehensive and contextually-based evaluation.

Patient exposure to particles occurs across a size continuum from submicron to millimeters, while risk assessments are generally conducted on particles reported as visible or subvisible components and not the aggregated risk of both subgroups. Assessing the potential risk that patient exposure to particles might present requires identifying the source of the particle; determining particle-specific attributes such as size, number, and composition; and ascertaining the intended use including the route and frequency of administration and relevant patient population characteristics ***(9).*** Any impact to the final product's benefit-risk profile that might result from the presence of particulate matter arises from the combination of these elements and calculated once the reference product and characteristics of the intended recipient are known.

In all instances of injections, the size of the particle is a restricting attribute. The particle must be small enough (or nonrigid) to pass through the bore of the needle (i.e., the particle must be less than the internal diameter of the needle in at least two dimensions). Detection probability increases with size such that larger particles are more likely to be identified with a detection probability (under ideal conditions) typically exceeding 95% for particles that are 200 µm and larger ***(2).*** Consequently, the product is either rejected during manufacture or found during dose preparation and discarded. There have been no reports identifying patient harm from randomly administered visible particulate matter in pharmaceutical preparations ***(8).*** Therefore, particles that are ultimately injected will be at or below the limit of detection of the qualified manufacturing environment with most of those particles being subvisible.

In most instances, the greatest risk to patients receiving injectable medicines is related to a lack of product sterility. For this reason alone, a robust inspection program is not sufficient; but rather, a broad

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range of other visible attributes is assessed along with laboratory testing of microbiological, chemical, and physical parameters. These are based on statistical sampling plans to assess the risk of elevated bioburden or other related harms in the batch. The risks to product sterility may be mitigated through processing decisions such as qualified component and product terminal sterilization processes (e.g., moist or dry heat, irradiation, or gaseous or vapor treatments). While the presence of a single particle is unlikely to provide a sufficient substrate to support microbiologic proliferation, multiple units identified with extrinsic particles should raise an alarm since the source is likely from a nonsterile environment. Appropriately qualified depyrogenation and autoclaving processes preclude the introduction of bioburden through primary packaging components.

Inert particles such as those from elastomers or glass generally pose little concern when introduced into a patient's subcutaneous tissue or muscle because they are typically sterile or sanitized. However, the particles could potentially stimulate a local immunologic reaction consisting of macrophages and foreign body giant cells at the interface of the particulate material (e.g., a granuloma), but it is likely that such a reaction would remain asymptomatic ***(10).*** More commonly, complications from subcutaneous and intramuscular medications arise from the irritating properties of the drug product and are often drug-specific but generally have a minimal impact ***(11).***

Intravenous (IV) infusion of particles may induce phlebitis due to the particles causing direct traumatic irritation to the vein or from chemical damage by undissolved particles ***(12).*** Recent observational studies have also described instances in select patient cohorts where morbidity has been reduced through the use of infusion filters, suggesting that particles in the infusate were responsible for the difference in outcomes ***(13).*** Other studies have failed to corroborate these findings in additional patient populations ***(14).***

A particle introduced intravenously that exceeds the internal diameter of a pulmonary capillary (7–12 µm) will lodge in that capillary or a more proximal pulmonary arteriole. While clinical outcomes are generally benign, such matter introduced into the systemic circulation might stimulate a local immunologic reaction (e.g., a granuloma) in the lung ***(15).*** These isolated, small parenchymal lesions generally remain asymptomatic. In addition, as demonstrated in animal studies, the massive infusion of particulates has been accompanied by histologic evidence of injury to pulmonary capillary endothelial cells ***(16),*** microscopic thrombi in the pulmonary capillaries ***(17),*** pulmonary microscopic granulomata ***(18),*** and hepatic inflammatory effects ***(19).*** An important distinction to note is that most of the studies above infused massive amounts of subvisible particles. While an all-inclusive review of the clinical implications of particulate infusions is beyond the intended scope of this technical report, the potential exists for the intravascular infusion of particulate matter to cause pathophysiologic injury depending upon the size and quantity of the particles, and, therefore, should be included as an element in any clinical risk assessment.

In summary, particulate matter will be present in every injectable dose. The risk of visible particles in injections to a patient is low from products manufactured and inspected under appropriately qualified current Good Manufacturing Practices conditions ***(20,21).*** Their origin, composition, size, numbers, and characteristics of the intended patient population are all important factors to weigh when assessing the clinical import from exposure.

3.2 Existing Test Methods and Limitations

Visual inspection of primary packaging components prior to use is hindered by four factors: the lack of a standardized industry inspection method for particle detection; the diversity of potential particle defects introduced through the manufacturing process; decreased visual detection capabilities on isolated or empty components; and the probabilistic nature of particle detection ***(3).*** Visual inspection of the packaging components is generally conducted by sampling and not inspection of every piece (100% inspection), thereby limiting the information necessary to construct an adequate library of the type and number of particles the process produces. Detection of particles by MVI, SAVI, or AVI is a

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cumulative function of the particles' visible attributes including particle size, shape, color, density, and reflectivity. Particle detection can also be more difficult in the empty components because the particles cannot be put in motion as with particles in a liquid product ***(22).***

The absence of standardized methods or visible particle acceptance criteria leaves it to each supplier and pharmaceutical manufacturer to establish these requirements for each product supplied. While regulatory guidance documents and industry publications may provide useful perspectives, the absence of a clearly articulated standard results in manufacturers subsequently adopting a hybrid approach using various portions of the published pharmacopeial ***(6,23)*** and International Standards Organization (ISO) methods ***(24).*** Identifying the limitations of test methods adapted for the evaluation of primary packaging components provides the opportunity to develop component-specific methods that would serve as a common standard for supplier-customer quality consideration.

Understanding the particle attributes is essential for each manufacturer when developing and adopting appropriate test methods, characterizing particle defect attributes, defining achievable detection capabilities, and assessing the potential impact of particles to the product's benefit-risk profile. The use of standard forensic methods of particle characterization ***(22)*** can be beneficial to inform the clinical risk assessment and direct allocation of resources to critical visual inspection and process control strategies. Additionally, the forensic datasets are important components to identify potential sources of particle ingress or generation, variability in raw material components, and monitor the stability of manufacturing processes ***(4).***

3.2.1 ISO Test Method 8871-3

ISO 8871-3 *Elastomeric parts for parenterals and for devices for pharmaceutical use — Part 3: Determination of released-particle count* ***(24)*** provides analytical methods for the extraction of loose particulates from the surface of elastomeric components used in pharmaceutical packaging applications. The ISO standard is split into subvisible particle extraction using light extinction (or light obscuration (LO)) and visible particle extraction followed by membrane filtration and microscopic examination (MM) for particle detection. Once particle extraction is complete, quantification is performed with an optical microscope and a calibrated reticle at approximately 50x magnification. Particle counts can be conducted either manually or via a validated automated microscopic method. Both the particle extraction and the counting technique are considered optimized in evaluating elastomer-associated visible particles. However, the method would require some level of enhancement for visible particles (i.e., particles >100 µm and fibers >300 µm as defined in this report) through the design of a reticle capable of differentiating particles from fibers and identifying a shaker for use in Europe that is comparable to that specified in ISO 8871-3. Additionally, further concern exists regarding the sensitivity of the method based on the surface area/number of stoppers sampled. Currently, there is no similar ISO method for particle extraction or characterization for parenteral container inspection.

3.2.2 U.S. Pharmacopeia Methods

USP Chapters <788> and <790> are generally used to provide analytical method guidance for packaging component particle assessment. USP <788> *Particulate Matter in Injections* ***(23)*** provides for two test methods for the detection of particles ≥ 10 µm in size: LO and MM. The methods do not appear suitable for elastomeric component particles since surface particle extraction methods are not outlined, and the sample preparation and counting techniques do not appear applicable for particles >100 µm. Likewise, the applicability of these methods in the evaluation of glass-related components hold limited utility for visible particles. While nonfixed particles in glass components might be identified using these methods, the diversity of particle morphology and container design significantly impact the sensitivity and applicability of these methods to glass components.

USP <790> *Visible Particulates in Injections* ***(6)*** became official in 2014 Chapter <790>, along with Chapter <1790> *Visual Inspection of Injections* ***(1)*** established requirements and provides supporting information to be used along with 100% inspection during the manufacturing process to demonstrate that the process has produced a batch that is "essentially free" of visible particulates.

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Table 3.2.2-1 provides a comparison of current inspection conditions and acceptance criteria for visible particles contained in the USP, EP, and JP. The chapters describe the manual inspection environment (lighting, background, and handling methods) that many component manufacturers have adapted for the inspection of empty containers and loose components. However, particle detection can also be more difficult in these components because the particles cannot be put in motion as with particles in a liquid product and the broad range of potential manufacturing defects generated by glass primary packaging suppliers. Additionally, these chapters are intended for filled-finished product and do not include defined test methods with clear requirements and specifications directly applicable to the individual packaging components used to manufacture injectable products. Therefore, the need for packaging component visual inspection guidance, which is currently lacking in the pharmacopeias, is critical to help characterize risk associated with these components and their impact on overall product quality assurance.

Table 3.2.2-1 Inspection Conditions and Acceptance Criteria for Visible Particles Found in the USP, EP, and JP.

Inspection Conditions	USP <790>	EP 2.9.20	JP 6.06
Illumination Intensity (lux)	2,000 – 3,750	2,000 – 3,750	2,000 – 3,750 8,000 – 10,000*
Inspection Time (sec)	10 (5 white, 5 black)	10 (5 white, 5 black)	10 (5 white, 5 black)
Backgrounds	black and white	black and white	black and white
Acceptance Criteria	"essentially free from visible particulates" ANSI/ASQ Z1.4 AQL=0.65%	"clear and practically particle-free"	"free of readily detectable foreign insoluble matter"

* Higher illumination intensity recommended for plastic containers.

3.2.3 European Pharmacopeia Methods

Two EP chapters govern subvisible and visible particles in sterile injectable products; EP 2.9.19 *Particulate contamination: sub-visible particles* ***(25)*** and EP 2.9.20 *Particulate contamination: visible particles* ***(26),*** respectively. The chapter on subvisible particles is harmonized with similar chapters found in the USP and JP, describing the same methods and containing the same acceptance criteria.

Chapter 2.9.20 describes the same inspection conditions as found in the USP and JP but does not provide a numeric acceptance criterion.

3.2.4 Japanese Pharmacopeia Methods

Like the USP and EP, the JP provides two chapters to address subvisible and visible particles. These are JP 6.07 *Insoluble Particulate Matter Test for Injections* ***(27)*** for subvisible particles and 6.06 *Foreign Insoluble Matter Test for Injections* ***(28)*** for visible particles. The subvisible particle test is also harmonized with the USP and EP while the test for visible particles has harmonized inspection conditions, again without a numeric acceptance criterion.

3.3 Conclusion

The observations documented in this report are intended to result in a standard test method. Such standardization is needed to produce reliable and comparable test results. The benefits of such a standard include clearer communication between supplier and users of pharmaceutical packaging components. It also permits setting meaningful limits and acceptance criteria for receipt and use of these important packaging components. Such limits would better align with the expectations for the finished products they are intended to protect and ultimately lead to more reliable supply of particle free sterile medicinal products.

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4.0 Elastomeric Closures Test Method

To standardized a method that will allow both elastomer suppliers and pharmaceutical companies to have meaningful data where it is reasonable to make comparisons, a new method was developed leveraging accepted aspects of the existing USP <788> Method 2 ***(23)*** and the ISO 8871-3 methods ***(24).*** The new method described in **Section 4.1** (the Enhanced Closures Testing Method) is specific to characterizing the visible particle load present in elastomeric closures and establishes a test to perform specifically on elastomeric closures after they are collected in their final packaging to ensure test results are representative of the product as used by pharmaceutical manufacturers.

4.1 Enhanced Closures Testing Method Development

Currently**,** specific but different guidance requirements exist for visible and subvisible particulates in parenterals. For final product inspection, the distinction between visible and subvisible is established by what can be seen by the unaided human eye when performing the compendial method inspection of approximately 10 sec per container in front of black and white backgrounds under controlled lighting. If a particle is observed in a final-filled container by the unaided eye of a qualified inspector, it is classified as a visible particulate defect and considered against the visible particle requirements such as those defined in USP <790> ***(6).***

In the case of packaging components for injectable products, suppliers of these components often perform an inspection for particulate matter at the time of release. Likewise, pharmaceutical manufacturing companies often perform a particulate test at incoming receipt. The testing of these components is intended to demonstrate that they meet specification and are appropriate for their intended use. As with final product testing, component testing should be capable of classifying any particulate on the components as either visible or subvisible. In cases where the packaging components are inspected by the unaided human eye for particulate, the same principles for classification apply. Namely, any particulate observed by the unaided human eye as part of packaging component testing will be considered visible.

In some cases, testing for particulate in packaging components will use analytical methods that exceed the ability of the unaided human eye. The most common example of such testing is a method that involves rinsing packaging components, filtering this rinsate, and then examining the filter under a microscope to measure and count any observed particulates. Microscopic examination will detect a wide range of particles including some which are not visible to the unaided human eye (i.e., subvisible) in addition to particles that are visible to the unaided eye. For this reason, the microscope confounds the ability of the test method to make the important distinction of visible versus subvisible. This challenge is common to any analytical or inspection method that exceeds human detection capability. Examples include current AVI systems, artificial intelligence applications, and numerous chemical and microbial assays.

4.1.1 Size Threshold

To allow for differences between visible and subvisible particles when using instrumental methods with capability that exceeds the unaided human eye, a size threshold is required to determine a visible particulate. The particulate size which is visible during inspection in filled product containers is the basis for the size thresholds. This approach ensures that particulate classification (i.e., visible and subvisible) in packaging components will more clearly connect to the regulations for visible and subvisible particulates in filled product containers.

Multiple studies ***(2)*** have been performed on the detectability of particulate matter of various sizes in the final containers of parenteral products when inspected by qualified inspectors following the compendial methods ***(6,26,28).*** These studies have demonstrated that under controlled conditions, the smallest size standard sphere that could be reliably detected (i.e., average PoD >70%) is 150 µm ***(2)***. The same studies demonstrated negligible detection of 50 µm particulates and variable detection for 100 µm particulates.

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Based on this understanding of detection as a function of particle size in parenterals, the value of 100 µm was selected as the threshold above which particles would be classified as visible when using an analytical method that exceeds the capability of the unaided human eye. The 100 µm threshold was conservatively selected to ensure that most particulate matter, even those with a moderate chance of being seen by the unaided human eye in a filled container, would be classified as "visible" by the analytical method using this threshold value. The 100 µm threshold is considered conservative because it is derived from the best case inspection conditions (i.e., clear solution, clear 10 mL vial, compendial lighting, background, and inspection time) and because it is lower than the 150 µm value established as the lower limit where detection in final containers is considered reliable (i.e., average PoD >70%).

When establishing a threshold defining a particle as visible, the particle geometry must also be considered. The 100 µm threshold was derived from studies of the detectability of standard spheres in liquid filled containers. Therefore, the 100 µm value is most applicable for an equant particulate which has an approximate 1:1:1 aspect ratio (length:width:thickness). This is similar to the ratio of a circle when considering the profile of the particulate in two dimensions. However, fibrous particles can have much higher aspect ratios. Such fiber-like particles need to be much longer in length as compared to an equivalent spherical diameter to be reliably detected in final product containers ***(1).*** The lower ability to detect a fiber-like particle versus that of a spherical particle is related to the two-dimensional profile of the two shapes and how this profile presents to the inspector. For example, a single 100 µm sphere has nearly eight times the two-dimensional surface area as a 100 µm fiber-like particle with an aspect ratio of 10:1 (length:width).

Based on the reduced detectability of fiber-like particles, a second threshold size to classify a particulate as visible was established for particulate with fiber-like geometries (i.e., those with a length to width aspect ratios >5). The size threshold to consider a fiber-like particulate visible was derived by calculating the length of a fiber with an equivalent two-dimensional surface area as the 150 µm spheres which were established as the lower limit where detection in final containers is considered reliable. For this calculation, an aspect ratio of 5 was used as a conservative limit such that higher aspect ratios would result in the classification of even longer fiber lengths as subvisible. Based on this approach, a particulate with fiber-like geometry would need to be greater than 300 µm to be considered visible when observed using analytical methods more capable than the unaided human eye. **Table 4.1.1-1** shows the two size limits that define a particle classified as visible or subvisible when using analytical methods to test primary packaging components for particulates.

The threshold values shown in **Table 4.1.1-1** apply to the testing of packaging components for particulates when an analytical method is used that exceeds the capability of the unaided human eye. These thresholds should not be applied to drug product final-filled container inspection, where the unaided human eye is the standard method used to assess whether visible particle levels meet the "essentially free" requirement and inspection performance may vary from the thresholds established here. Specifically, finished product inspection performance may be affected by the container color and design and the product type (e.g., solution color, opacity, density). These confounding characteristics are removed from the test described herein. Additionally, the size thresholds should not be applied to any particulate that is detected by the unaided human eye during testing of the packaging components, as the act of detecting the particulate would be definitive proof that the particulate is visible. That classification should not be overridden by subsequent analytical characterization.

Table 4.1.1-1 Threshold Value to Use when Classifying Particles as Visible or Subvisible (by methods more sensitive as compared to the unaided eye)

Category	Aspect Ratio (length:width)	Visible Threshold	Subvisible
Particle	<5	100 µm	<100 µm
Fiber	≥5	300 µm	<300 µm

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4.1.2 Novel Particle Size Measurement Tool

The assessment of particles in elastomer components traditionally involves rinsing the components, then filtering the rinsate, and finally examining the filter under a microscope. No thresholds to differentiate visible from subvisible, as discussed above, have been established. To consistently apply the principles of the visible particle size thresholds defined in **Section 4.1**, a novel measurement tool was developed for use in elastomer particle testing involving microscopic analysis of the filtrate.

Figure 4.1.2-1 depicts a novel reticle design that enables an analyst to readily determine whether particles observed microscopically on a filter should be considered visible or subvisible. The reticle has two shapes in the view field: a circle, and a rectangle. The circle is to be sized such that for a given microscope under the intended magnification, the circle is equivalent to a diameter of 100 µm. Similarly, the rectangle is 300 µm × 60 µm to address both size and aspect ratio thresholds for fiber-like particles.

The reticle design shown in **Figure 4.1.2-1** allows for rapid and consistent application of the particle size and aspect ratio thresholds established for this test. When an analyst examines the filter through a microscope, they simply compare any particle observed against the rectangle and circle templates in the reticle. Any particle that fits entirely within either of the two shapes is considered subvisible and is not to be counted as visible. Those that extend beyond the boundaries of either template are counted as visible, and depending on the aspect ratio as a particle or fiber.

4.2 Summary for Performing the Enhanced Closures Testing Method

A summary of the Enhanced Closure Testing Method for counting visible particles on elastomeric closures follows. Refer to **8.1 Appendix A** for a detailed discussion on the development of this method.

A number of parts (normalized for surface area) are placed in an Erlenmeyer Flask, and 50 ml of surfactant solution is dispensed and added to the flask. The flask is then placed on an orbital shaker to agitate the parts and remove visible particles from the surface. After 20 seconds of shaking the flask is removed from the shaker and the surfactant solution is immediately filtered through a membrane filter (a gray filter was used to enhance contrast for both light and dark colored particles and fibers). The rinsing process is repeated and the filter is allowed to dry. Once the filter is dry, any visible particles present are counted by using a specialized reticle and an optical stereomicroscope.

A thorough discussion of the sample preparation process and the counting methodology used to evaluate within and between operator replicate samples can also be found in **8.2 Appendix B**.

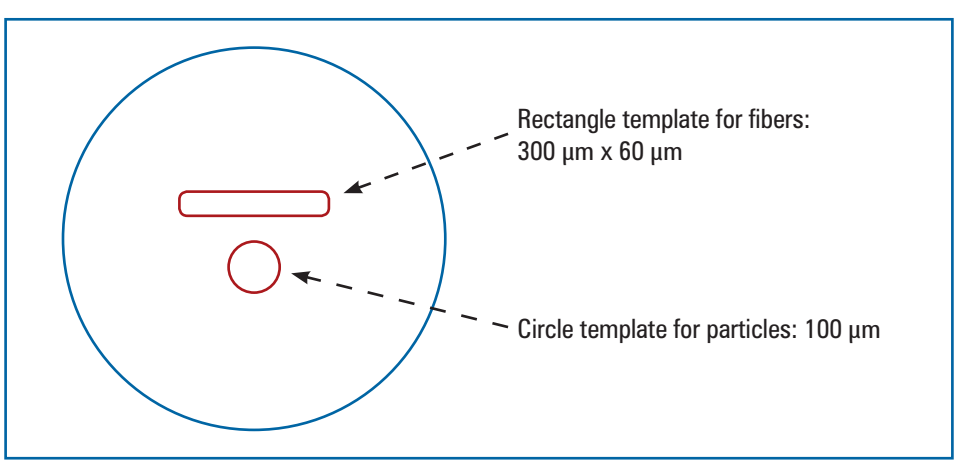

Figure 4.1.2-1 Microscope Reticle Design For Elastomer Component Particle Analysis

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4.3 Method Qualification Strategy and Lessons Learned

Preliminary studies for attribute agreement revealed that rigorous training and easy-to-follow instructions reduces the subjectivity of the analyst's counting technique. counting techniques. Detailed instructions and the use of the unique counting reticle described in **Section 4.1** are key to this training. Repeatability of the particle and fiber count data was proven when multiple analysts inspected each other's filter samples. The material found to be best suited for training analysts and qualification of the method was to select stopper populations prior to their pharmaceutical washing step. These stoppers provided particle counts that were sufficiently high to yield meaningful statistical evaluation.

In attempting to measure repeatability when performing multiple sample extractions from a single stopper bag or across multiple stopper bags, significant variability was observed. Specific factors contributing to this variability in the particle counts were identified through the multiple design of experiments (DOE) based investigations and subsequent review of the findings with the operators involved. The inherent lack of homogeneity of particle and fiber deposition on the surface of elastomeric components that are packaged within a bag or collectively transported in multiple cartons from the same batch was one of the underlying root causes for the inconsistent initial results observed early in the development of this test method. Repetitive handling, sampling, and shipping greatly increases variability and localization of particles.

Another challenge identified during development and qualification of this method was to determine the number of stoppers needed for training and qualification. Because it is common for commercial stopper batches to have particle counts near zero, these batches are not ideal to train analysts or qualify the method. There were several attempts made to purposely seed particles into stopper batches, but these were not successful. Generating appropriate particulate loading through various attempts at artificially seeding with shredded fragments of rubber and aliquots of a cellulose emulsion proved to be ineffective due to inconsistent distribution within the bag, which in turn resulted in highly variable recovery.

Additional detail on the development of this method and the evaluation of contributions to method variability can be found in **8.1 Appendix A**.

5.0 Glass Container Inspection Method

Today, no alignment exists within the pharmaceutical industry on a method to inspect empty glass containers. This is of special importance for RTU containers since they will not undergo any further cleaning procedure(s). Inspection of empty glass containers may be performed by the glass manufacturer/converter or by the pharmaceutical company prior to use. It may be performed on all units or a statistically valid sample collected from a larger batch.

5.1 Inspection of Ready-to-Use Empty Glass Containers

The proposed visual inspection method which follows was based largely on the method and inspection conditions defined in the EP ***(26)***, USP ***(6)***, and JP ***(28)*** for filled containers. **Table 5.1-1** summarizes the key requirements established to improve the performance of the method employed in the study.

The procedure is a manual inspection method where the quality decision to either accept or reject the container is made by a trained person. MVI is a probabilistic process where detection rates less than 100% are to be expected ***(3,29,30)***. To understand the probabilistic nature of this type of inspection a comprehensive study was performed using the defined method to inspect defect kits of empty glass containers seeded with different types of particles of varying size and location on and in the container.

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Table 5.1-1 Key Attributes of Proposed Inspection Conditions and Method

Attribute	Requirement
Inspectors	• Pass a medical eye examination before beginning visual inspection activities. Near vision acuity should be equivalent or better than 20/20 ft (or 6/6 meters) or equivalent. Inspectors should be retested and meet this criterion at least every two years. • Use of optical correction such as prescription glasses is optional if needed to achieve normal vision. • Avoid direct viewing of the light source. • Take regular breaks to avoid fatigue and eye strain. Do not exceed one hour of continuous inspection.
Training and Qualification	• Train inspectors on the specific method and materials used. • Qualify inspectors with test sets with representative defect examples and good containers. • Define and document qualification requirements. • Evaluate inspector performance at least every two years.
Inspection Environment	• Perform inspection in a clean and quiet area. • Reduce ambient lighting. • The use of gloves and gowning are recommended.
Equipment	• An inspection workstation which provides a light intensity between 2,000 and 3,750 lux. • Black and white backgrounds.
Method	• Perform inspection without magnification. • Inspect containers for approximately 5 sec against each background. • Hold containers approximately 30 cm from the eye. • Hold containers 10-20 cm from the light source. • Rotate container to examine all visible areas.

This was necessary to better understand the capability of the method for empty containers and to establish criteria for inspector qualification. **Section 5.3** outlines the results of this cross-industry study.

5.1.1 Operator Training for the Enhanced Container Inspection Method

To achieve reliable and consistent results, inspectors must be trained on the inspection method before performing routine inspection and their inspection performance verified by testing both initially and then every two years (minimum) thereafter. This requalification interval is less frequent than that for inspectors assessing filled units but is considered a practical interval since empty containers are reassessed when filled both by 100% and a subsequent acceptance sampling (AQL) inspection. The qualification testing should be performed using seeded test kits containing both good (blank) and bad (defective) samples. Clear acceptance requirements should be defined against which the inspectors are evaluated. Limits on the number of missed defects and good containers incorrectly rejected (false rejects) are necessary when setting the acceptance criteria.

5.1.2 Equipment and Setup

The results of manual inspection are highly influenced by the light intensity in the inspection zone. Light intensity between 2,000-3,750 lux at 10 cm from the light source is recommended based on current finished product inspection conditions found in the USP, EP, and JP. The light intensity should be confirmed prior to use and the intensity recorded at the point of inspection.

Container samples should be examined against both black and white backgrounds to maximize the contrast between the glass and the defect of interest. A matte, nonglossy background is recommended to avoid interference from glare or reflection.

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Figure 5.1.2-1 shows a typical visual inspection workstation with diffuse top lighting and black and white backgrounds. A light table may also be used to provide diffuse light upwards to and through the object under inspection to supplement the diffuse illumination from above. Eight of the eleven companies engaged in the study used the light table in addition to the standard inspection workstation.

Figure 5.1.2-2 shows a container being inspected in the workstation using the recommended method and conditions. **Figure 5.1.2-3** provides additional detail on the location of the container under inspection relative to the light source and viewing position of the inspector.

Figure 5.1.2-1 Visual Inspection Workstation

Figure 5.1.2-2 Example Of Container Inspection Using Recommended Method And Relative Placement In The Inspection Hood

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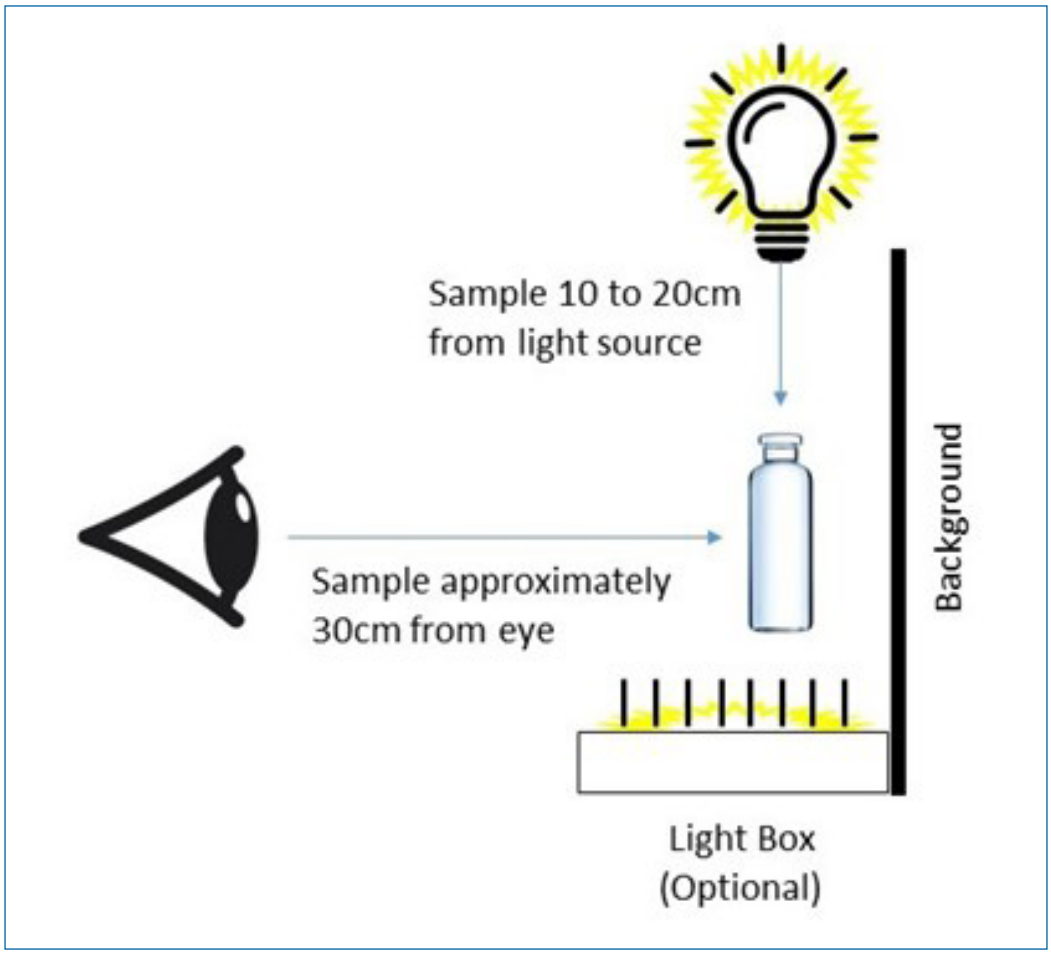

Figure 5.1.2-3 Placement Of Container Within The Inspection Hood During Inspection

The work area should be neatly organized. For example, samples for inspection may be placed on the left of the inspection area and two empty trays placed on the right: one for passing and one for failing units. Trays should always be clearly labeled and preferably color coded. Discard all samples good and bad after the inspection and any follow-up investigation completed.

5.1.3 Magnification Requirements To Inspect Glass Containers

Magnification, other than that for optical correction to normal vision, is prohibited for routine release inspection. This is recommended to keep alignment with the pharmacopeial inspection conditions. USP General Chapter <790> ***(6)***] states that inspection should be performed without magnification. However, USP Information Chapter <1790> (***1***) adds that while "not recommended for use during routine inspections, magnification can be helpful for critical examination of a small number of units, as may be needed during an investigation". If magnification is used for investigational purposes, strict guidance should be provided to avoid it becoming an uncontrolled part of the routine inspection process.

5.1.4 Glass Container Inspection Procedure

Detailed instructions for empty glass container inspection can be found in **8.4 Appendix D**. Information recorded during or after inspection may include details such as the type and location of any particles observed, and any other defects observed. On completion of the inspection, the data should be evaluated against the established acceptance criteria to determine the disposition of the batch. During inspection, the operator should avoid direct viewing of the light source and take regular eye breaks to avoid eye strain which may adversely influence the results of the inspection. Inspection time should be based on the parameters defined in the qualification.

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5.2 Method Qualification Strategy and Detection Capability

To set realistic targets for human inspection performance, the detection capability of the method being used for inspection must be understood. A significant portion of the work reported here centered on performing inspections on seeded containers assembled into test kits to better assess the detection capability of the method described.

For all observations reported in this study, the light intensity was confirmed as ≥3,000 lux at the point of verification which is at the upper end of the inspection conditions defined in the EP ***(26),*** USP ***(6),*** and JP ***(28),*** of 2,000-3,750 lux.

Three equivalent test kits were created representing a wide range of visible particles types and sizes commonly found in glass containers. Specific examples were based on survey results ***(31)*** and the experience of glass team members. Eight different particulate types (glass, metal, cellulose, hair, plastic, and rubber [black, grey, and orange]), were defined and used when preparing the kits. The kits were prepared without any visible adhesive which might bias the measured detection rate. Each container was checked for particles prior to seeding, each unit was seeded with a single particle and then sealed to prevent further contamination. For these studies, each kit contained 60 each of vials, syringes, and cartridges, making a total of 180 pieces. The kits were compared and checked against each other to ensure that they were of similar composition. The locations of seeded particles are shown in **Figure 5.2-1** and the type and nominal size of particles are detailed in **Table 5.2-1**. Only seeded containers were used in these kits to assess differences in ability to detect particles when compared with others of varying type and size.

The sample kits were inspected by five pharmaceutical companies and six tubular glass convertors. Each of the organizations taking part in the study had the kits examined by five inspectors, six times each for a total of thirty inspections per company. To better blind the inspection results, the inspections were not performed consecutively. A total of 59,400 independent observations were made. However, some data points were excluded from the study because they did not meet the stated inspection requirements (e.g., the inspection time was out of the specified range). Ultimately, 49,360 valid data points were used to analyze the performance of the inspectors and the inspection method.

Particle size measurement. Because of the unique optical properties and geometric configuration of both containers and particles, different methods were used to determine the nominal size of seeded

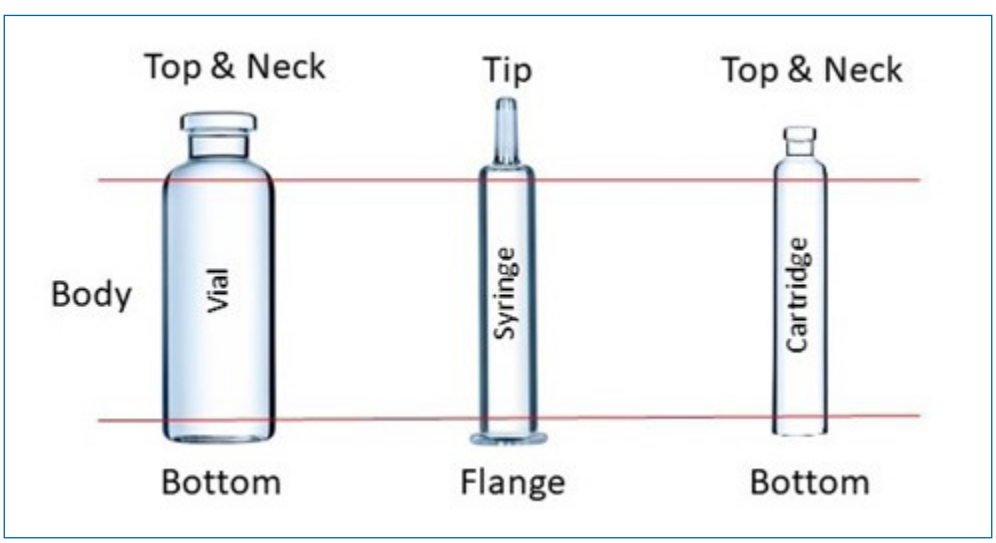

Figure 5.2-1 Locations Where Particles Were Seeded Into Or Onto Containers (Vials, Cartridges, And Syringes) In Prepared Test Kits

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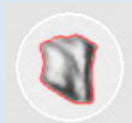
Table 5.2-1 Summary of Nominal Particle Size by Material

Type	Material	Color	Photo	Particle/Fiber Size (µm)						
				100	200	300	500	700	800	1,000
Particles	Glass	Transparent		X	X	X				
	Rubber	Orange – Cartridges Black – Syringes Gray – Vials (shown)		X	X	X	X			
	Metal	Silver/Gray		X	X	X				
Fibers	Cellulose	White				X	X	X		X
	Plastic	White				X	X	X	X	
	Hair	Brown				X	X			

particles. **Table 5.2-2** shows the methods used for the various materials incorporated into the test kits. These were chosen to best represent how these particles appear to the human eye during inspection.

Length is simply a measure of the distance along the longest axis. Feret Max is the maximum distance between parallel tangents at opposing particle borders and Rect A, the longer side of a rectangle constructed around the particle which is in contact with the two points used to define Feret Max as shown in **Figure 5.2-2.**

Table 5.2-2 Measurement Method Used to Determine Nominal Particle Size

Particle Type	Key Dimension (Sizing Parameter Used)
Glass	Feret Max
Metal	Feret Max
Cellulose	Length
Hair (Brown)	Length
Plastic (From Tub)	Rect A
Rubber (Black)	Rect A
Rubber (Grey)	Rect A
Rubber (Orange)	Rect A

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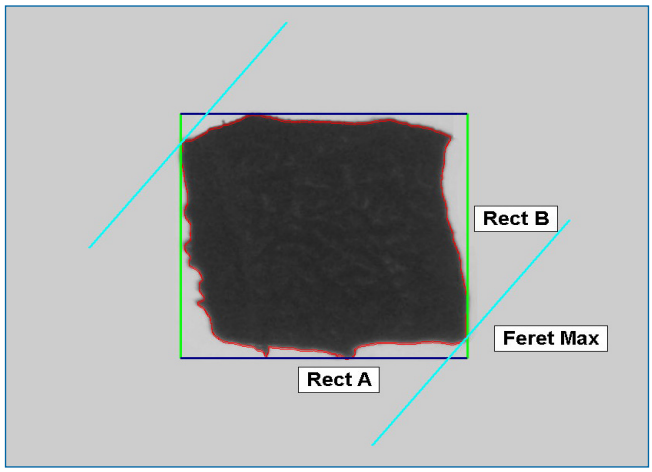

Figure 5.2-2 Examples of Feret Max And Rect A And B Particle Size Measurements.

5.3 Study Design

A total of eleven companies participated in the study: five from the pharmaceutical industry and six manufactures of the primary packaging materials. Each company used five inspectors for the study, each inspector performed the inspection six times, producing a total of 59,400 data points. The data was reviewed and confirmed as valid prior to being used in the statistical analysis.

Based on the inspection results, non-fibrous particles of 125 µm in size had an average PoD of approximately 70%. The PoD increases to 95% for a particle of 300 µm. **Figure 5.3-1** shows inspection performance for both particles and fibers and confirms past studies with filled containers ***(2)*** showing increasing PoD with increasing particle and fiber size. The diameter of the circles at specific x-y locations is proportional to the number of data points which occurred at that location. For fibers, the PoD is 70% for 300 µm fibers and increases to 90% for fibers between 800-1,000 µm in length. This difference in detectability can be shown to be largely due to the aspect ratio. Fibers such as hair and plastics (tub residues) were more easily detected while cellulose fibers were harder to detect.

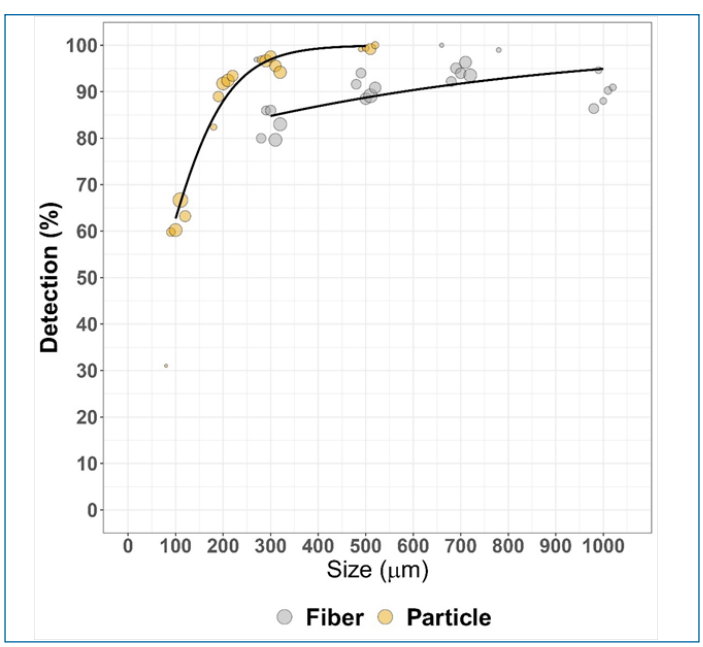

Figure 5.3-1 Probability Of Detection For Particles And Fibers Of Varying Size During Inspection Of Empty Glass Containers (Circle diameter is proportional to the number of data points falling at a specific x-y location)

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Analysis of study results further discussed in **8.3 Appendix C** also found that variability between inspectors from different companies was negligible, suggesting this method should provide robust results across the industry.

5.3.1 Normalizing the Particle Size

To better understand the difference in detectability between fibers and particles the impact of the aspect ratio on detectability (which is the relationship between the width and length of the fiber) must be considered. To address the impact of the aspect ratio the data was normalized by multiplying the Key Dimension of each particle by its SF. This conversion was applied to both particles and fibers to create the approximate area that would be seen by the inspector's unaided eye.

$$\text{Normalized Particle Size} = \text{Key Dimension} \times \text{Shape Factor (SF)}$$

Using this transformation, the PoD for fibers and particles is similar to the PoD seen in **Figure 5.3.1-1** The combined PoD then becomes 80% for 100 µm particulate and 95% for particulates of 200 µm. The diameter of the circles at specific x-y locations is proportional to the number of data points which occurred at that location.

In summary, a core element of the study was the creation of sample kits containing realistic defect examples where the PoD was not compromised or biased by the sample preparation (e.g., if the glue holding the particle is visible). By preparing these kits with defects that appear as those found during routine inspection the performance of the inspector using this method could be verified in a controlled way.

The study produced an extensive database with 49,360 validated data points, thereby providing a high degree of statistical confidence in the conclusions drawn on inspection performance. This study found that using the compendial inspection conditions the PoD for visible particles ranged from 70% for

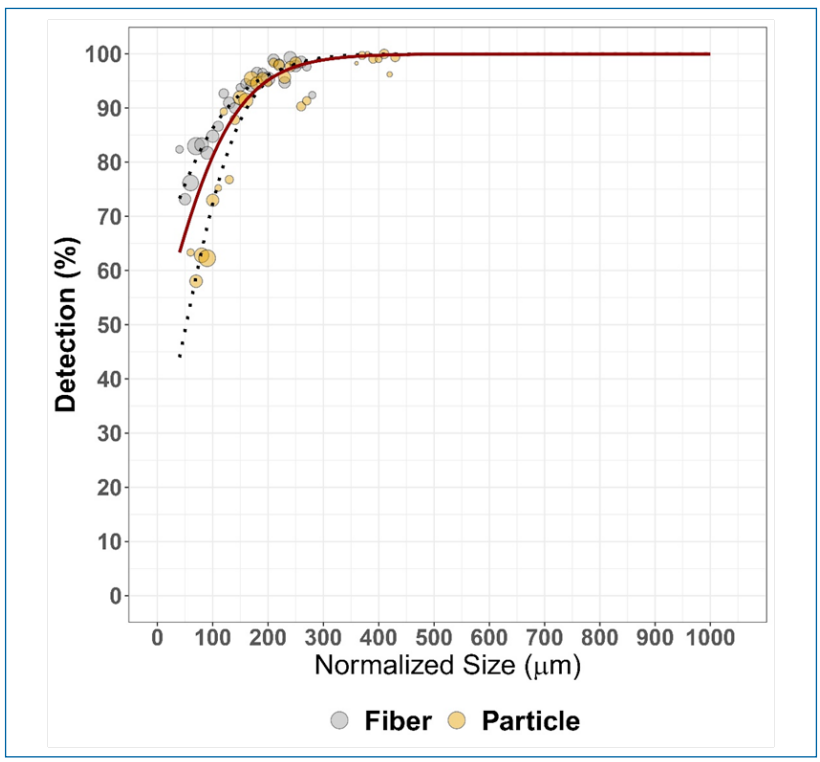

Figure 5.3.1-1 Probability Of Detection For Particles And Fibers Of Varying Size During Inspection Of Empty Glass Containers With Size Normalized To Apparent Area (Circle diameter is proportional to the number of data points falling at a specific x-y location)

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125 µm particles to 95% for particles >300 µm. Due to their aspect ratios, fibers are more difficult to detect and their corresponding PoD are 80% for 300 µm fibers to 90% for fibers >800 µm.

These results align well with detectability studies performed on filled containers following the compendial methods ***(6,26,28).*** These earlier studies demonstrated that under controlled conditions, the smallest size standard sphere that could be reliably detected (i.e., average PoD >70%) was 150 µm.

In conclusion, from this study with empty containers and the previous work on filled containers, this method can be used successfully for the inspection of unfilled RTU containers by both the pharmaceutical manufacturing companies and the glass component suppliers. Inspectors using this method should be trained and qualified using test kits with well characterized and representative samples that accurately simulate routine defects and defect free samples. Additional review and discussion of the data analysis on inspection performance can be found in **Section 8.3 Appendix C**.

6.0 Conclusion

Visual inspection continues to be an important step in the manufacture and delivery of safe and effective injectable medicines. Current inspection methods, manual and machine assisted, can provide reliable monitoring and control of the visual characteristics of injectable products, including visible particles. The work described in this technical report is intended to further the development of methods to increase reliable supply and reduced rejection of injectable drug products. Reliable detection and control of visible particles in and on the components used to contain these products is essential to meeting this goal.

This technical report provides an improved method to assess visible particle burden on elastomeric closures. This method uses much that is known and available but provides standardization of key parameters including extraction, agitation, and size thresholds (100 µm for particles and 300 µm for fibers) to aid in consistent enumeration. This is expected to assist both the manufacturers of such closures and the pharmaceutical companies who use them in packaging their drug products. A standard method to assess visible particles will hopefully lead to a practical working standard which incorporates risk-based limits suitable for the intended application.

This report also describes a study to better assess the ability to inspect empty glass containers. While several studies have looked at PoD in finished, filled containers, little has been done with empty containers to-date. This study shows similar inspection performance between empty and filled containers and supports the use of standard compendial visual inspection conditions to perform inspection of empty glass containers. An average PoD of 70% was achieved for 125 µm particles and

300 µm fibers. These agree with and support the thresholds chosen for the Enhanced Closure Testing Method. Further analysis of the study data shows that PoD of particles and fibers can be related to each other through assessment of the apparent visible area.

Continued advancement in visual inspection and particle control will require broad industry support. Ongoing engagement with additional companies and regulatory organizations would further the work that was described herein and lead to acceptance of common methods and acceptance limits for visible particles on or in primary packaging components.

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8.0 Appendices

8.1 Appendix A. Additional Information on Enhanced Closures Testing Method Development

Test methods that yield variable data with great certainty, such as dimensional measurements or potency, can be simply qualified or validated. However, human performance, such as visual inspection, is not directly measurable or validated. As such, the strategy for qualifying a methodology based on attribute data required an initial verification of the sample preparation process and the counting method that would ultimately be used to evaluate the comparability of within operator replicates or between operators and cohorts of samples. To accomplish this, the evaluation and testing strategy was separated into two phases, pilot (method development) and qualification. These are outlined in **Figure A1.**

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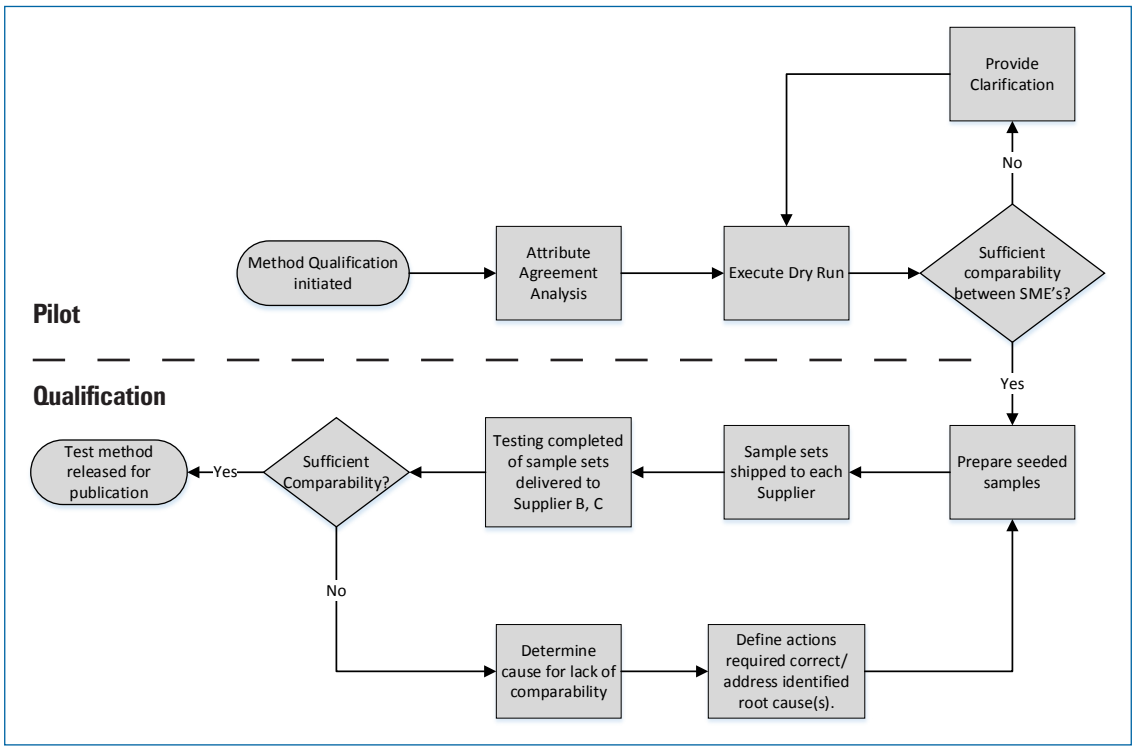

Figure A1 Process Map of the Pilot (method development) and Qualification Strategy

The results from the pilot phase highlighted the necessity of seeding samples prior to distribution and evaluation by the three participating suppliers. Therefore, the formal qualification study was stratified as outlined in **Figure A2**. Three seeding groups were created by adding particles of elastomeric material and dissolved fiber. Although the fibers were added using a defined volume of a prepared aqueous mixture, the particle generation process employed a novel technique of creating fragments of elastomeric material and manual deposition onto the surface of each stopper sample contained in the sample bags. The resulting groups of sample bags were then provided to all three suppliers with the goal of minimizing the impact of variability of the three seeding groups.

Statistical analysis of the resulting particulate count data revealed that the use of a nonstandard sample extraction and transfer process (due to the inherent nature of the elastic particulate matter being extracted) of the seeded particles and fibers may have contributed to unintended consequences for the counting process. Inconsistencies in the seeding process and the possible impact of forces exerted upon the sample bags during transit seemed to be contributing factors that ultimately yielded the failing results observed. Significantly, counting results from all operators were repeatible when cross comparisons were performed between operators.

An extensive investigation was performed to quantitatively evaluate the impact of the multiple sources of varibility that were contributing to the intial results and define viable means of mitigation. Based on learnings from the previous studies, the team chose to discontinue the use of artificially seeded particles and opted, instead, to use stoppers that were not processed in a pharmaceutical washer in which a higher native particle burden was exhibited. Accordingly, a final conclusive study was designed and executed to both evaluate the impact of transportation on the range of particles and fibers observed and to serve as the basis for recommending the specific method described in this report. To limit

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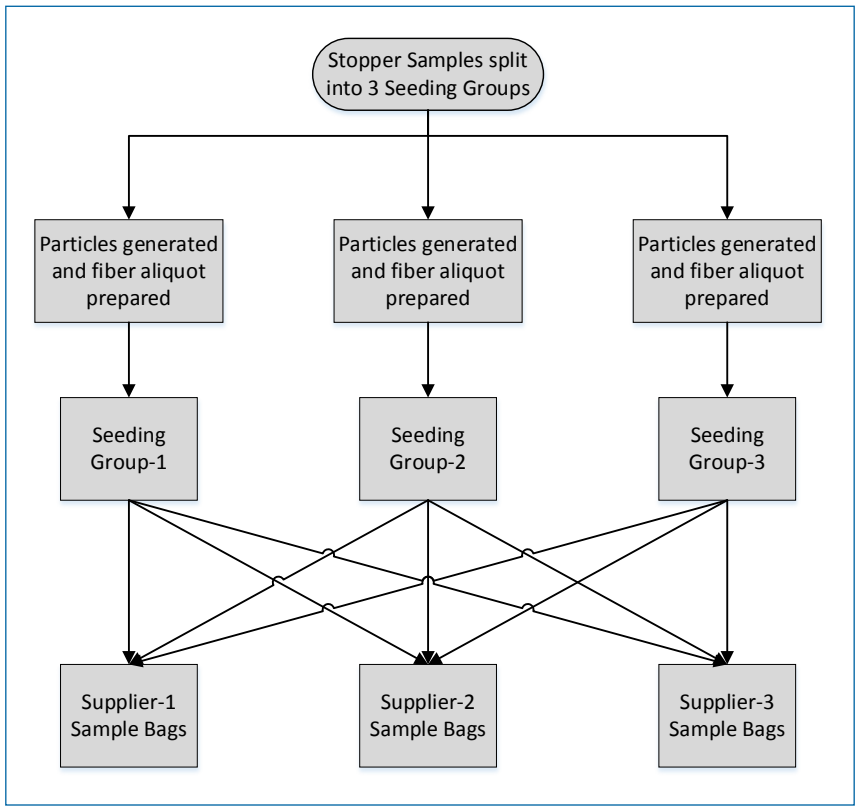

Figure A2 Method Qualification Sample Generation Stratification

the impact of nonhomgeniety in particle deposition when multiple smaller sample sets are created and then divided into one or more cohorts, a single bag of unseeded, non-washed stoppers was used between all three suppliers in series. Although multiple operators performed the replicate extractions, all individuals sampled from the previously mentioned single bag. The process map for this study is illustrated in **Figure A3** below.

Overall Study Results

The particle and fiber count data from each supplier was statistically evaluated for reproducibility between operators and suppliers (using an Analysis of Variance (ANOVA) hypothesis test) and repeatability of counting output between operators using a paired t-test when evaluating the same filter samples. Handling and time difference between inspections were minimized in all cases. An evaluation of the comparability of method results for a group of inspectors was completed for Supplier 1 and the results are shown in **Figure A4 through Figure A6**.

Figure A4 shows the results for the particles >100 µm. Comparability of extraction was proven between three operators with ANOVA p-value = 0.590. As shown in **Figure A5**, reproducibility of the counting method for particles >100 µm was proven between three operators with paired t-test with resulting p-values of 0.078 – 0.761. The calculated p-value from the ANOVA results for fibers >300 µm did not reveal comparability of extraction between the three operators for Supplier 1, and the confidence interval for the fiber results extended into negative values in **Figure A6**. Nonetheless, reproducible results were achieved as demonstrated by the outcome of paired t-tests between Operator 1 vs Operator 3 and Operator 2 vs Operator 3 shown in **Figure A7.**

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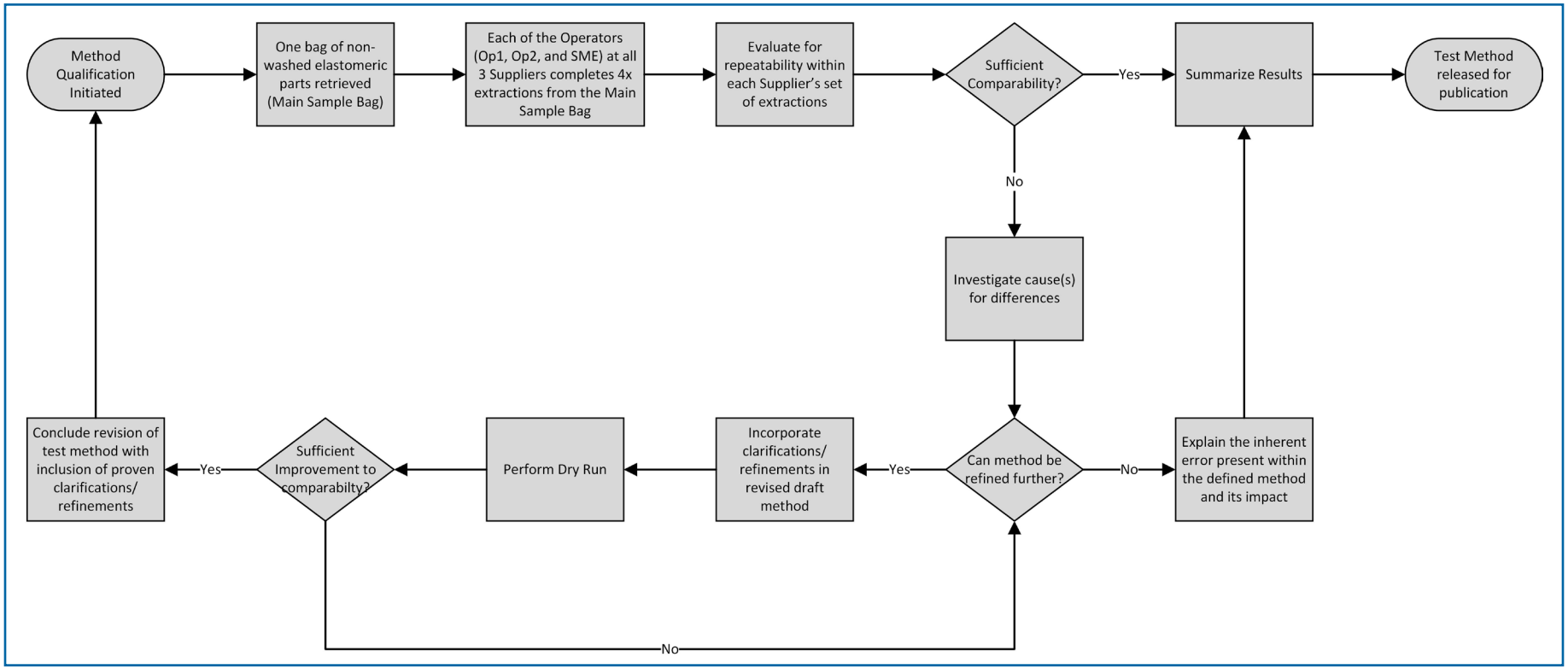

Figure A3 Process Map of Final Method Qualification Study

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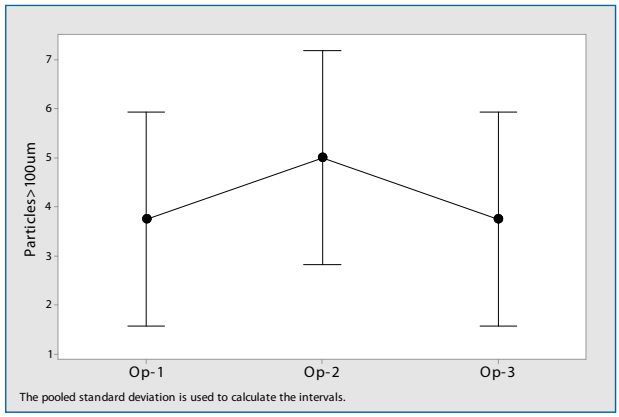
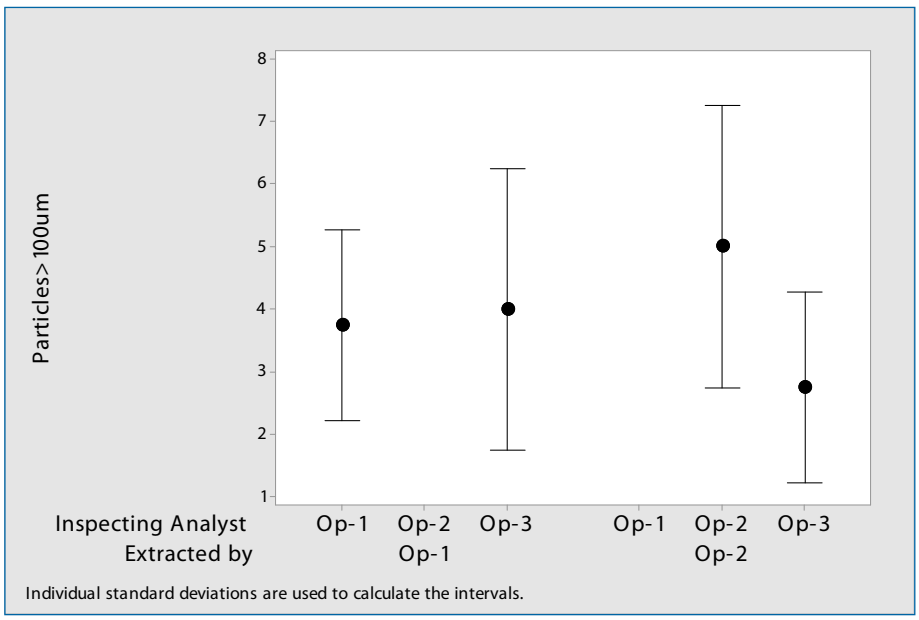


Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Extraction by _2	2	4.167	2.083	0.56	0.590
Error	9	33.500	3.722		
Total	11	37.667			

Figure A4 Interval plot of Supplier 1 results for particles >100 µm for each operator



Paired T-test for Op 1 vs Op3

Test

Null hypothesis	H_0: µ_difference = 0
Alternative hypothesis	H_1: µ_difference ≠ 0

T-Value	P-Value
0.33	0.761

Paired T-test for Op 2 vs Op 3

Test

Null hypothesis	H_0: µ_difference = 0
Alternative hypothesis	H_1: µ_difference ≠ 0

T-Value	P-Value
-2.63	0.078

Figure A5 Interval plot of replicate inspection results for Supplier 1, Operators 1 vs 3 and Operator 2 vs 3, for particles >100 µm

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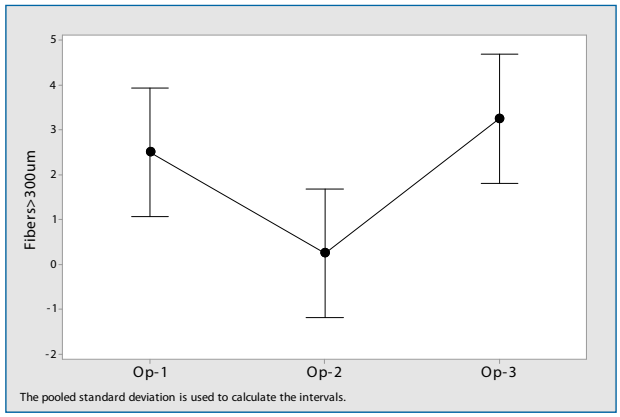
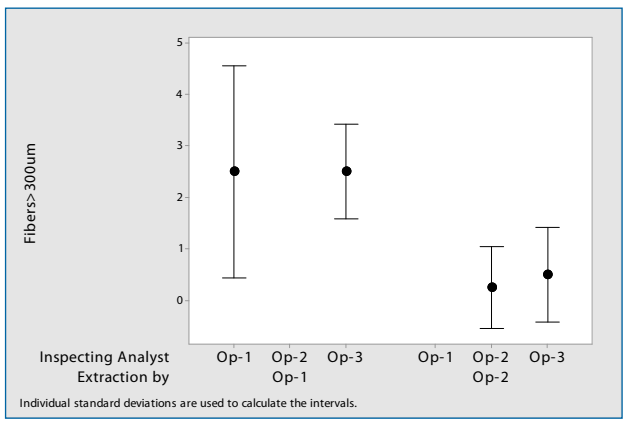
A similar evaluation of the comparability of method results for a group of inspectors for Supplier 2 was also completed and the results are shown in **Figure A8 through Figure A10**. **Figures A8 and A9** show results for the particles >100 µm and **Figure A10 through Figure A11** show results for fibers >300 µm. Although calculated p-values from the ANOVA results would appear to indicate comparability of extraction between the three operators, these results are not feasible since the 95% confidence intervals extend to negative counts. The reproducibility of the counting method for particles >100µm (shown in **A9**) was demonstrated between the three operators producing paired t-test with resulting p-values of 0.444-0.903. **Figure A11** shows the results for the fibers >300 µm. Reproducibility of the counting method was demonstrated between all three operators with paired t-test results yielding a p-value =1.000.



Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Extraction by _5	2	19.50	9.750	6.05	0.022
Error	9	14.50	1.611		
Total	11	34.00			

Figure A6 Interval plot of Supplier 1 results for particles >300 µm for each operator



Paired T-test for Op 1 vs Op3

Test

Null hypothesis	H_0: µ_difference = 0
Alternative hypothesis	H_1: µ_difference ≠ 0

T-Value	P-Value
0.00	1.000

Paired T-test for Op 2 vs Op 3

Test

Null hypothesis	H_0: µ_difference = 0
Alternative hypothesis	H_1: µ_difference ≠ 0

T-Value	P-Value
1.00	0.391

Figure A7 Interval plot of replicate inspection results for Supplier 1, Operators 1 vs 3 & Operators 2 vs 3, for fibers >300 µm

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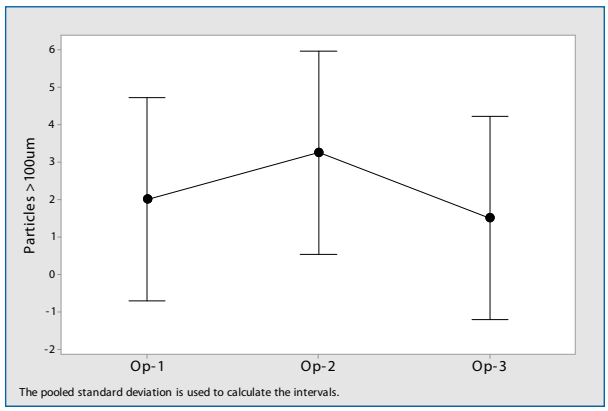
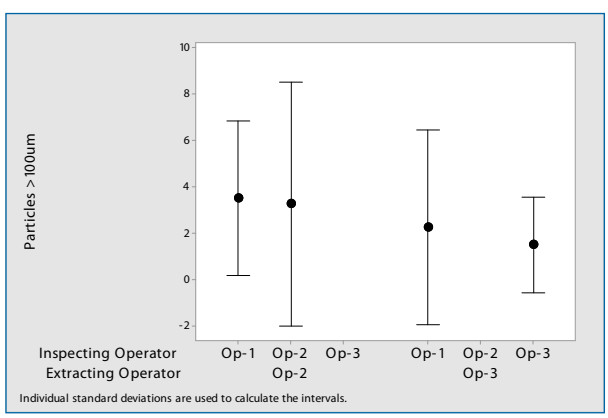


Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Inspecting Operator	2	6.500	3.250	0.57	0.587
Error	9	51.750	5.750		
Total	11	58.250			

Figure A8 Interval plot of Supplier 2 results for particles >100 µm for each operator



Paired T-test for Op 1 vs Op2

Test

Null hypothesis	H_0: µ_difference = 0
Alternative hypothesis	H_1: µ_difference ≠ 0

T-Value	P-Value
0.13	0.903

Paired T-test for Op 1 vs Op 3

Test

Null hypothesis	H_0: µ_difference = 0
Alternative hypothesis	H_1: µ_difference ≠ 0

T-Value	P-Value
0.88	0.444

Figure A9 Interval plot of Supplier 2 group performance for particles >100 µm

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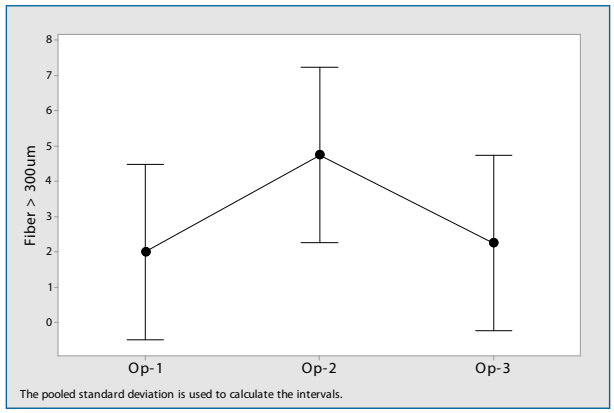
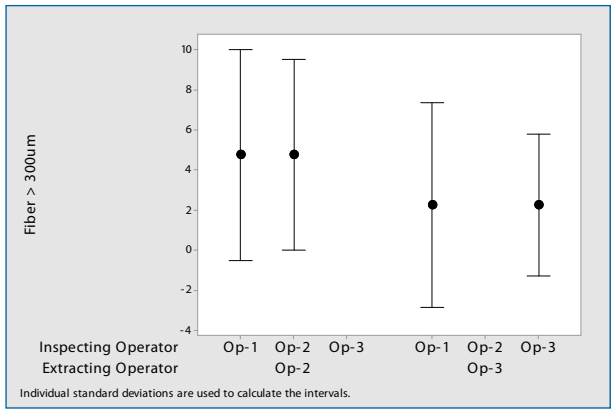


Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Inspecting Operator	2	18.50	9.250	1.91	0.203
Error	9	43.50	4.833		
Total	11	62.00			

Figure A10 Interval plot of Supplier 2 results for fibers >300 µm for each operator



Paired T-test for Op 1 vs Op2

Test

Null hypothesis	H_0: µ_difference = 0
Alternative hypothesis	H_1: µ_difference ≠ 0

T-Value	P-Value
0.00	1.000

Paired T-test for Op 1 vs Op 3

Test

Null hypothesis	H_0: µ_difference = 0
Alternative hypothesis	H_1: µ_difference ≠ 0

T-Value	P-Value
0.00	1.000

Figure A11 Interval plot of Supplier 2 group performance for fibers >300 µm

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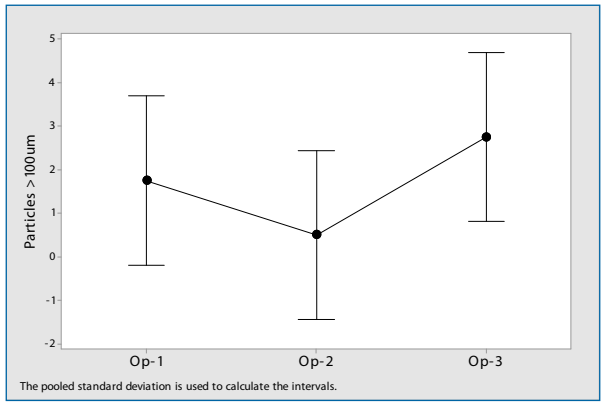
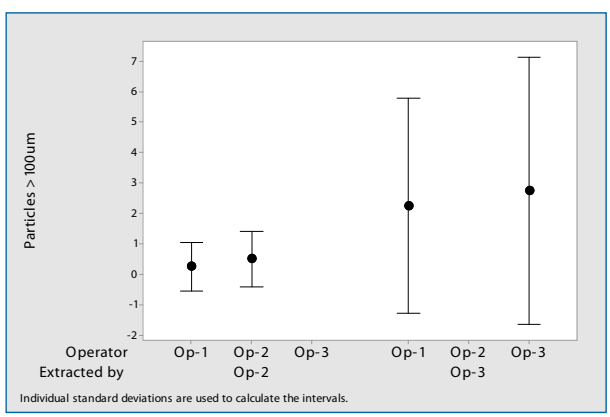
Finally, a further evaluation of results from inspectors at Supplier 3 is shown in **Figure A12 through Figure A15**. **Figure A12 and Figure A13** show the results for the particles >100 µm. Although calculated p-values would appear to indicate comparability of extraction between three operators with ANOVA, these results are again not feasible since the 95% CI's extend to negative counts. The reproducibility of the counting method was confirmed between the three operators using a paired t-test resulting in p-values of 0.391-0.604. **Figure A14 and Figure A15** show the results for the fibers >300 µm. Comparability of extraction was not confirmed between the three operators with a resulting ANOVA p-value of 0.006. Reproducibility of the counting method was, however, proven between these three operators using a paired t-test with resulting p-values of 0.391-1.000.



Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Inspecting Operator	2	10.17	5.083	1.73	0.232
Error	9	26.50	2.944		
Total	11	36.67			

Figure A12 Interval plot of Supplier 3 results for particles >100 µm for each operator



Paired T-test for Op 1 vs Op2

Test

Null hypothesis	H_0: µ_difference = 0
Alternative hypothesis	H_1: µ_difference ≠ 0

T-Value	P-Value
-1.00	0.391

Paired T-test for Op 1 vs Op 3

Test

Null hypothesis	H_0: µ_difference = 0
Alternative hypothesis	H_1: µ_difference ≠ 0

T-Value	P-Value
-0.58	0.604

Figure A13 Interval plot of Supplier 3 group performance for particles >100 µm

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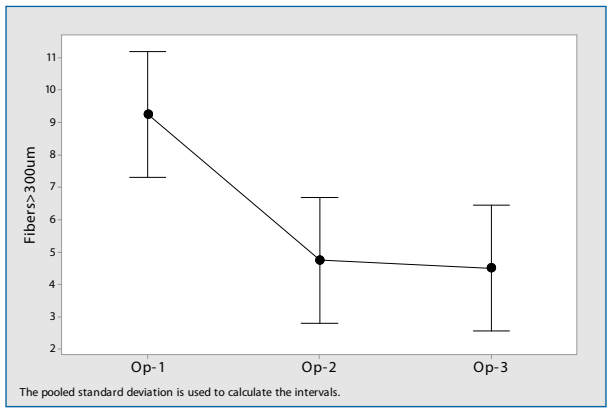
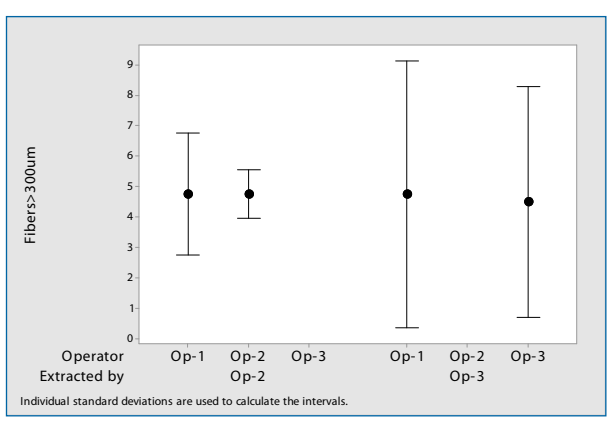


Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Inspecting Operator	2	57.17	28.523	9.71	0.006
Error	9	26.50	2.944		
Total	11	83.67			

Figure A14 Interval plot of Supplier 3 results for fibers >300 µm for each operator



Paired T-test for Op 1 vs Op2

Test

Null hypothesis	H_0: µ_difference = 0
Alternative hypothesis	H_1: µ_difference ≠ 0

T-Value	P-Value
0.00	1.000

Paired T-test for Op 1 vs Op 3

Test

Null hypothesis	H_0: µ_difference = 0
Alternative hypothesis	H_1: µ_difference ≠ 0

T-Value	P-Value
1.00	0.391

Figure A 15 Interval plot of Supplier 3 group performance for fibers >300 µm

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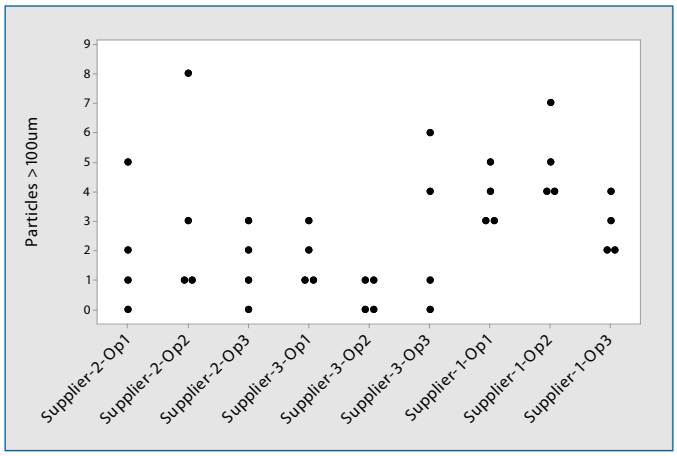
An evaluation of the comparability of method results across the three separate suppliers where testing was performed is shown in **Figures A16 through A19**. **Figure A16 and Figure A17** show the results for particles >100 µm. When the individual operators' results are aggregated into supplier groups the values fail to prove comparability with a resulting p-value of 0.022. **Figure A16 and Figure A17** show the results for fibers >300 µm. The same scenario as that observed for particles holds true for fibers where the values fail to prove comparability with a resulting p-value of 0.000.



One-way ANOVA: Particles > 100 µm versus Supplier

Method

Null hypothesis	All means are equal
Alternative hypothesis	Not all means are equal
Significance level	$\alpha = 0.05$

Equal variances were assumed for the analysis

Factor Information

Factor	Levels	Values
Supplier	3	Aptar, Datwyler, West

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Inspecting Operator	2	30.17	15.083	4.27	0.022
Error	33	116.58	3.533		
Total	35	146.75			

Figure A16 Comparison of inspection performance by both supplier and operator for particles >100 µm

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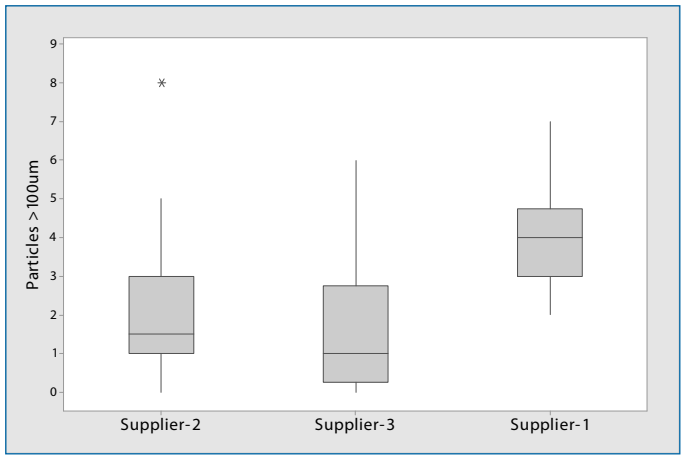


Model Summary

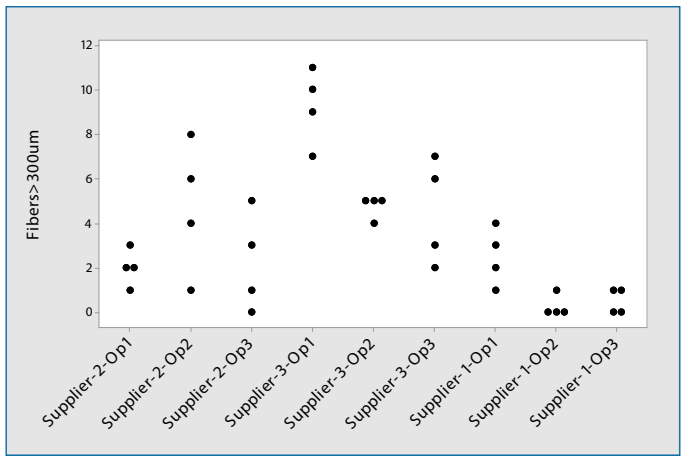
S	R-sq	R-sq(adj)	R-sq(pred)
1.87958	20.56%	15.74%	5.46%

Means

Supplier_1	N	Mean	StDev	95% CI
Supplier_1	12	3.833	1.403	(2.729, 4.937)
Supplier_2	12	2.250	2.301	(1.146, 3.354)
Supplier_3	12	1.667	1.826	(0.563, 2.771)

Pooled StDev = 1.87958

Figure A17 Box plot of performance by supplier for particles >100 µm



One-way ANOVA: Fibers > 300 µm versus Supplier

Method

Null hypothesis	All means are equal
Alternative hypothesis	Not all means are equal
Significance level	$\alpha = 0.05$

Equal variances were assumed for the analysis

Factor Information

Factor	Levels	Values
Supplier	3	Aptar, Datwyler, West

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Inspecting Operator	2	158.2	79.083	15.86	0.000
Error	33	164.6	4.967		
Total	35	322.8			

Figure A18 Comparison of inspection performance by both supplier and operator for fibers >300 µm

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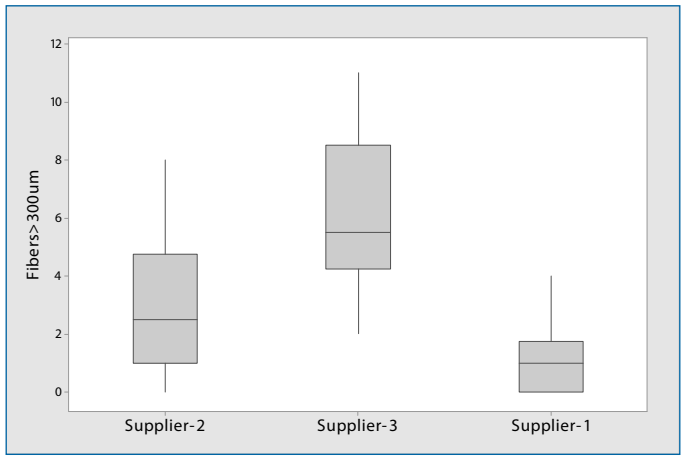

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
2.23324	49.01%	45.92%	39.31%

Means

Supplier_1	N	Mean	StDev	95% CI
Supplier_1	12	1.083	1.311	(-0.228, 2.395)
Supplier_2	12	3.000	2.374	(1.688, 4.312)
Supplier_3	12	6.167	2.758	(4.855, 7.478)

Pooled StDev = 2.23324

Figure A19 Box plot of performance by supplier for fibers >300 µm

Conclusion. The original strategy for confirming the robustness of the test method for extracting and counting the particulate present on the resulting membrane filter was intended to demonstrate quantitatively the repeatability and reproducibility of both components of the test method. Preliminary studies for attribute agreement revealed that rigorous training and easy to follow instructions are key to reducing the subjectivity of the analyst's counting techniques. Once this foundation of objectivity was formed for operators, the validity of their results was not in question. Repeatability of the particle and fiber count data was confirmed not only when extracted samples were independently reinspected, but also when all operators inspected each other's filter samples. This is evident in the hypothesis test results (well above a p-value of 0.05) that were obtained throughout the studies and the multiple qualification cohorts. On the other hand, reproducibility of output data with replicate extractions or between each operator's sample sets was not easily demonstrated.

Through multiple investigatory design-of-experiments and review of the findings with the operators involved, specific factors that contributed to providing what was initially thought to be inexplicable variability in the particle counts were identified. The inherent lack of homogeneity of particle and fiber deposition on the surface of elastomeric components that are packaged within a bag or collectively transported in multiple cartons from the same batch was one of the underlying root causes for the inconsistencies of the initial data gathered. As a result, managing sample bag to sample bag variability proved to be one of the largest factors in obtaining sufficiently low CV values. Repetitive handling, sampling, and shipping greatly increases variability and localization of particles within the bags.

Results from the final transportation and qualification study demonstrated that seeding particles was not necessary to generate counts that are sufficiently high to yield meaningful statistical evaluation if unwashed samples are used instead of RTS product.

Although several studies revealed relatively good comparability within replicate extractions for individual operators within a supplier's testing team, consistent agreement within the range of values for particles or fibers was not demonstrated across all organizations. Counting technique for particles and

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fibers applied to sets of replicate extractions yielded good comparability between local operators for all three suppliers. This comparability between local operators for all three suppliers demonstrates that the newly developed method is a robust and repeatable procedure.

8.2 Appendix B. Method: Determination of Visible Particles and Fibers on Elastomeric Components by Membrane Filtration and Microscopic Examination

Introduction. The purpose of this method is to provide a procedure to quantify the number of visible particles present on elastomeric closures used for injectable pharmaceutical products. The procedure describes how the elastomeric closures are to be sampled from the fully assembled package, then rinsed to release any particles present, and subsequently sized and counted using a stereomicroscope.

Scope. This method is applicable to all RTU and RTS elastomeric closures but only applies to visible particles. The method requires that such closures be tested from fully assembled final packaging to reflect how they are received and used by the pharmaceutical manufacturer.

Fiber-like particles (i.e., length to width aspect ratio of ≥5) have significantly less surface area than a sphere of equivalent diameter to the fiber length. This results in significantly less detection capability for a 100 µm long fiber as compared to 100 µm diameter sphere. To ensure this analytical method is detecting visible particles only, fiber-like particles have been given a larger visible size threshold of 300 µm.

Equipment and materials.

Bolded items below are considered essential for executing the method.

1. **Laminar flow cabinet meeting the requirements of ISO Class 7 or better**
2. Dispensing pressure vessel
3. Filter jet dispenser
4. Jet spray nozzle
5. Analytical petri slide
6. Analytical filter holder
7. Grey membrane filter for counting, gridded, 25 mm, 0.8um porosity
8. Membrane filter for jet pistol, 25 mm
9. **Orbital shaker with a 12 mm arm length or equivalent. The suitable RPMs for orbital shaker arm lengths are show in** Table B1 **below. The RPM ranges are adjusted to ensure comparable performance between shakers of different arm lengths.**
10. Timer traceable to an internal standard
11. Wide-mouth Erlenmeyer flask, 300 mL
12. Graduated cylinder, class A to deliver, 50 mL
13. **Stereomicroscope capable of approximately 50x magnification**
14. **Use Reticle, Klarmann part number KR-32972-1 at 50x magnification (or equivalent).**
15. **Calibrated stage micrometer to verify reticle size at magnification used**

Table B1 Orbital Shaker RPM Ranges

Orbital Arm Length	Allowable RPM Range	
	Low	High
10 mm	355	380
12 mm	300	325
25 mm	200	225

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Equivalent equipment and materials are acceptable but must be demonstrated to be suitable for the intended use. Equipment must be qualified and calibrated prior to use.

Reagents

1. Surfactant, Polysorbate 80, reagent grade
2. Water, purified grade 1 or 2 as specified in ISO 3696 (32) and filtered through a terminal filter with NMT 1 um pore size.

Preparation for the test. Ensure that the environment in which the test is to be carried out is free from extraneous particles which could cause interference. This involves wearing suitable garments such as synthetic, low particle-shedding frocks, hair nets and beard guards and gloves within a suitable laminar flow cabinet. Tools and equipment must be cleaned before use by rinsing thoroughly with water (as defined above as Reagent 2). The exterior of sample bags should be cleaned prior to being introduced into the cleaned test environment. Cleanroom certified wipes should be used to clean the bags by wiping in overlapping straight lines, away from the operator.

Preparation and dispensing of 0.03% Polysorbate 80 solution. Dissolve 3g of Polysorbate 80 in 10L of water. Alternative volumes may be prepared for use by keeping this ratio constant. Prepared solution should be dispensed under pressure to permit in-line filtration with a filter with a pore size of NMT 1 µm.

Blank testing. Blanks should be tested before and after a sample series. Blanks must meet acceptance criteria prior to testing samples. If blank analysis after sample series do not meet acceptance criteria, sample results are considered invalid.

The use of the terms *clean* or *cleaned* below refer to tools that have been rinsed with clean and filtered water before use.

Blank testing procedure

1. Using cleaned forceps, rinse gridded, then un-gridded side of filter for 20 sec per side and place on filter holder.
2. Place clean funnel on filter holder.
3. Fill 50 mL of Polysorbate 80 solution (as defined as Reagent 1 above) in a clean graduated cylinder.
4. Pour the 50 mL of Polysorbate 80 solution into a clean Erlenmeyer flask or accurately add 50 mL of Polysorbate 80 solution directly to the Erlenmeyer flask through dosing.
5. Shake the flask on the orbital shaker at the appropriate rpm (see **Table B1**) for 20 sec.
6. Decant the solution from the flask into the funnel.
7. Repeat steps 3-6.
8. Rinse the inside of the Erlenmeyer flask thoroughly and pour Polysorbate 80 rinse into the funnel.
9. Rinse the inside of the funnel thoroughly.
10. Remove the funnel and with clean forceps, place the filter in a clean petri slide.
11. Label the petri slide appropriately.
12. Dry the filter by placing it in the laminar air stream with lid ajar at least until the filters have dried.

Blank testing acceptance criterion. Blanks are acceptable if no particles or fibers >100 µm are observed under the specified microscopic examination conditions.

Sample testing procedure. The use of the terms clean or cleaned below refer to tools that have been rinsed with clean and filtered water prior to use.

1. Using cleaned forceps, rinse unprinted, then printed side of filter for 20 sec per side and place on filter holder.

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2. Place clean funnel on filter holder.
3. Open the sample bag carefully and transfer 100 cm^2 of parts to a clean Erlenmeyer flask.
4. Fill 50 mL of Polysorbate 80 solution into a graduated cylinder.
5. Pour the 50 mL of Polysorbate 80 solution into a clean Erlenmeyer flask or accurately add 50 mL of Polysorbate 80 solution directly to the Erlenmeyer flask through dosing.
6. Shake the flask on the orbital shaker at the appropriate RPM for 20 sec.
7. Decant the solution from the flask into the funnel.
8. Repeat steps 4-7.
9. Discard stoppers.
10. Rinse the inside of the Erlenmeyer flask thoroughly with water (as defined above as Reagent 2) and pour rinsate into the funnel.
11. Rinse the inside of the funnel thoroughly with water (as defined above as Reagent 2).
12. Remove the funnel and with clean forceps, place the filter in a clean petri slide.
13. Dry the filter by placing in the laminar air stream with lid ajar at least until the filters have dried.

Filter Evaluation. The calibration of the stereomicroscope should be verified daily prior to use using a micrometer that is calibrated at the operational magnification. Use suitable, direct illumination (e.g., from a goose-neck lamp, light emitting diode [LED] light source recommended) with angles set between 0 and 10° from the vertical viewing axis. The magnification used for examining the filters should be approximately 50x. Once the lighting conditions are set, the same conditions should be used throughout the analysis. The focus can be adjusted throughout the analysis to keep the particles in focus.

Particles must exceed the borders of the templates in the reticle and meet the established aspect ratio requirements to be counted.

- Particles: >100 µm in length (to be determined using the circle template)
- Fibers: >300 µm in length (to be determined using the rectangle template)

The reticle described in **Section 4.1** and shown in **Figure B1** must be used to accurately classify any observed particles as visible or subvisible for either particle or fiber-like form. The following diagram provides instruction for use of the reticle.

Calculations. Use the following equation to calculate the number of visible particles per 10 cm^2 for both particles and fibers.

$$\frac{\textit{visible particles}}{10\ cm^2} = \frac{\textit{visible particles counted on filter} \times 10}{\textit{number of parts used} \times \textit{surface area per part}}$$

$$\frac{\textit{visible fibers}}{10\ cm^2} = \frac{\textit{visible fibers counted on filter} \times 10}{\textit{number of parts used} \times \textit{surface area per part}}$$

Reporting. Report the number of visible particles/10cm^2 and visible fibers/10cm^2

8.3 Appendix C. Additional Information on Empty Glass Inspection Capability Studies

Study Test Kits

The glass container inspection study required the test kits to be of sufficient quality so as not to bias the study results. This was achieved through the following steps.

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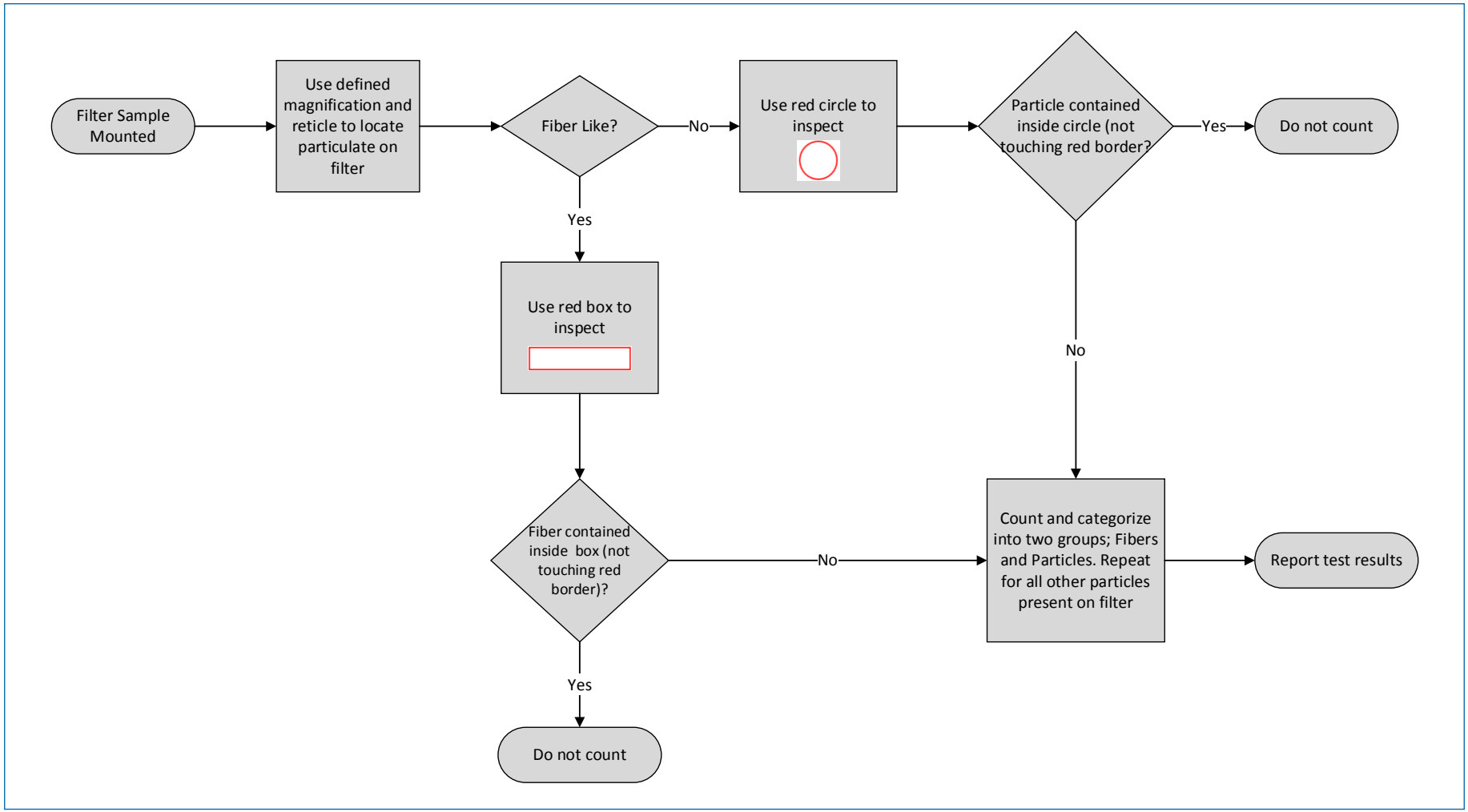

Figure B1 Flowchart showing the use of the specified reticle to determine status of any observed particles on the filter

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Table C1 Summary of Nominal Particle Size by Material

Type	Material	Color	Photo	Particle/Fiber Size (µm)						
				100	200	300	500	700	800	1,000
Particles	Glass	Transparent		X	X	X				
	Rubber	Orange – Cartridges Black – Syringes Gray – Vials (shown)		X	X	X	X			
	Metal	Silver/Gray		X	X	X				
Fibers	Cellulose	White				X	X	X		X
	Plastic	White				X	X	X	X	
	Hair	Brown				X	X			

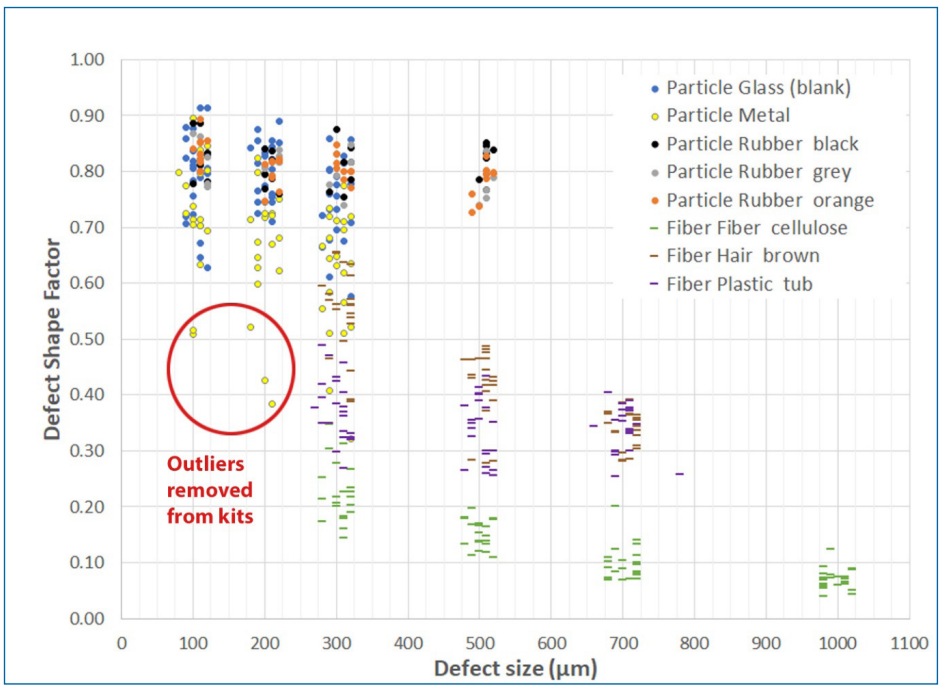

Figure C1 Shape factor vs size for the particles and fibers seeded into test vials

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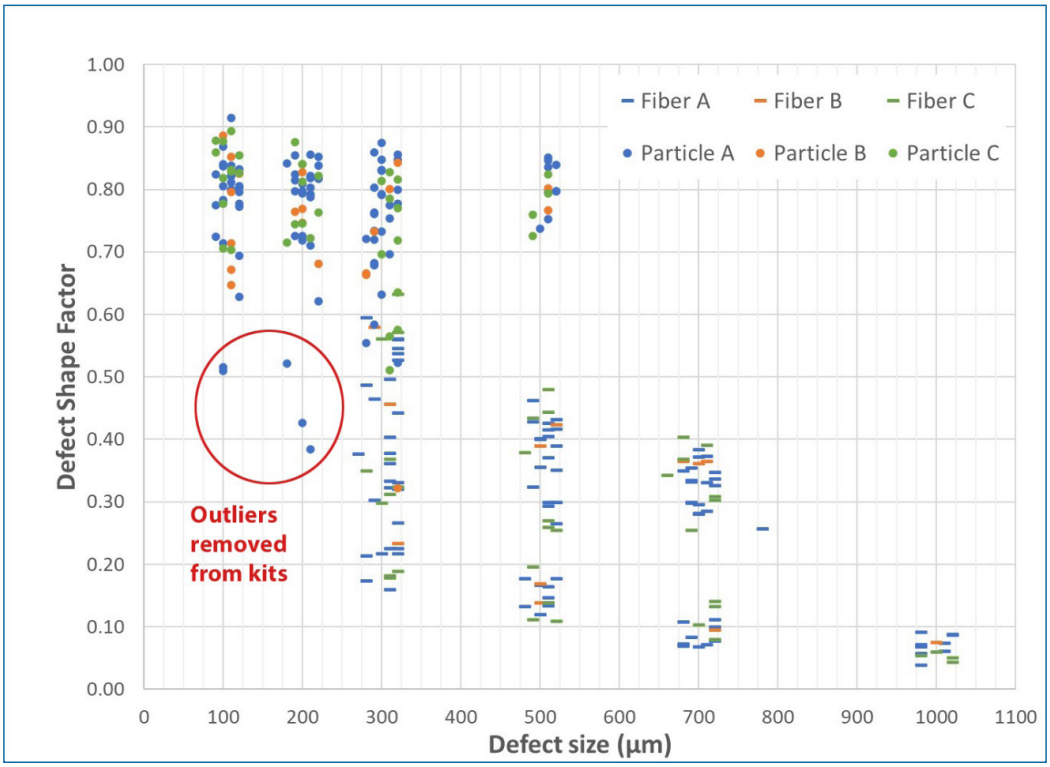

Figure C2 Graphic comparison of test kit composition (kits A, B, and C)

The containers were washed and seeded with a particle of known size and type. Each particle was digitally photographed and measured before being glued with a highly transparent, UV-cured glue. Each container was then sealed to prevent further particle contamination. Three equivalent kits were produced and used for this study to ensure that the results were not influenced by the composition of a specific kit. A critical analysis of the composition of each kit was performed before use. **Table C1** shows the intended particle types and size ranges while **Figure C1** shows the actual results. **Figure C2** provides a graphic comparison of the composition of the kits (A, B, and C) that were prepared and used in the study showing both SF and size of the particles and fibers seeded into test containers. The circled outliers were removed from the study kits to assure conformance with the intended kit design.

Data Collection

Study Data. A total of eleven companies participated in the study: five from the pharmaceutical industry and six manufactures of the primary packaging materials. Each company used five inspectors for the study; each inspector performed the inspection six times, producing a total of 59,400 data points. The data were reviewed and confirmed as valid prior to being used in the statistical analysis.

Inspection Rate. Sufficient time must be allowed for a thorough inspection of each container. The EP ***(26),*** USP ***(6),*** and JP ***(28)*** each specify a reference time of inspection per container of ten 10 sec; five sec each against black and white backgrounds. Before executing the study, an estimate was made for inspection of the complete test of 60 containers using the minimum inspection time defined by the compendia. The estimated target time to complete the inspection of the kit, allowing for the necessary container handling and inspection was determined to be 10-20 min.

On completion of the study, an analysis was performed to establish inspection time boundaries based on actual time used and parameters for determining invalid and missing data. The lower limit was set at 9 min while the upper limit was calculated as the median plus 1.66 standard deviations, resulting in an upper limit of 31min. Results obtained outside of this range were not considered valid and were not used in the analysis.

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Figure C3 shows that about 86% of the test runs were considered valid for the analysis based on inspection time. Data were also excluded if a container was broken or unavailable for inspection. This resulted in 49,360 of the 59,400 data points being used for subsequent analysis.

Main influences on probability of detection

Operator Experience. The inspectors participating in this study had differing levels of glass inspection experience. For pharmaceutical manufacturers, the inspectors typically inspect filled containers, whereas those at the suppliers primarily focus on empty containers. Overall, however, the results from both groups of inspectors were found to be comparable. See **Figure C4** for particle inspection results and **Figure C5** for results for fibers. Larger circles indicate a greater number of observations with the same value.

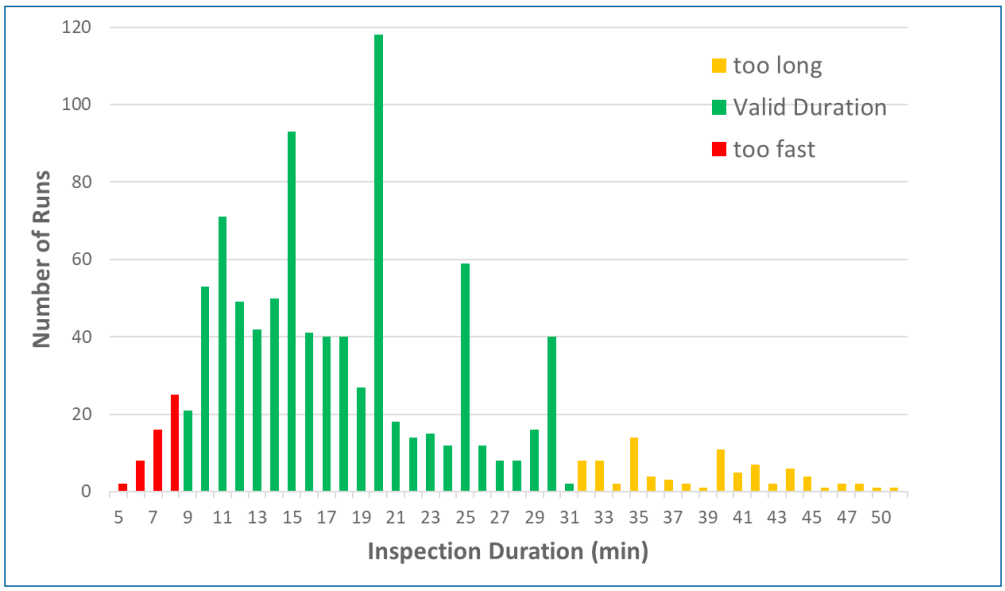

Figure C3 Distribution of time required to complete inspection of test kit

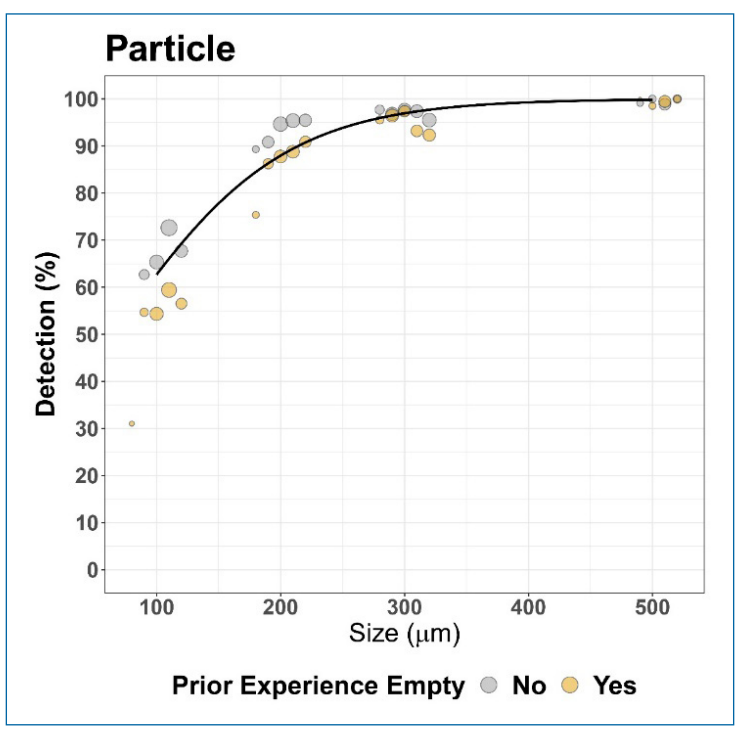

Figure C4 Particle inspection results with inspectors having prior experience with empty containers (Yes) and those without such experience (No). Circle diameter is proportional to the number of data points falling at a specific x-y location.

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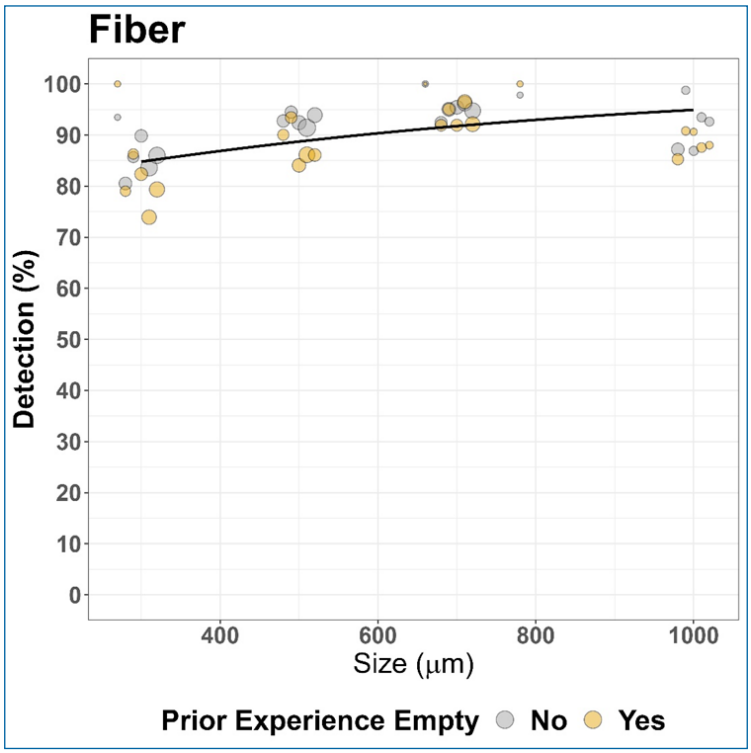

Figure C5 Fiber inspection results with inspectors having both prior experience (Yes) with empty containers and those without such experience (No). Circle diameter is proportional to the number of data points falling at a specific x-y location.

Particle/Fiber type. The material composition of the particle had a significant influence on the PoD observed. Glass, metal, and grey rubber particles (100 to 200 μm) were found to have a lower PoD than orange or black rubber particles of similar size as shown in **Figure C6**.

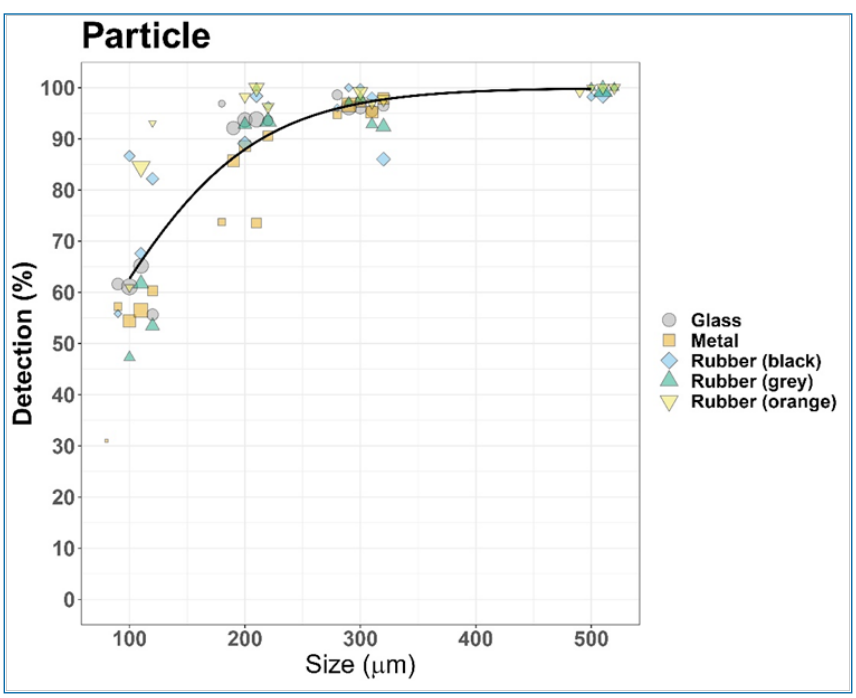

Figure C6 Probability of detection for particles of differing type and size. Circle diameter is proportional to the number of data points falling at a specific x-y location.

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This becomes more pronounced with fibers, where cellulose fibers have a statistically lower PoD than that for brown hair or white plastic as shown in **Figure C7**. Optical contrast is necessary for reliable detection and greatly impacts the resulting PoD.

Container Size/Type. **Figure C8** for particles and **Figure C9** for fibers show the influence of container size and type on inspection results. The vials used in the study were 30 ml and had a surface area approximately three times larger than either the 1 mL syringes or 3 mL cartridges used. This resulted in a decrease in the inspection time per unit surface area when examining containers of larger size and suggests a reason for lower detection rates in the vials used for this study.

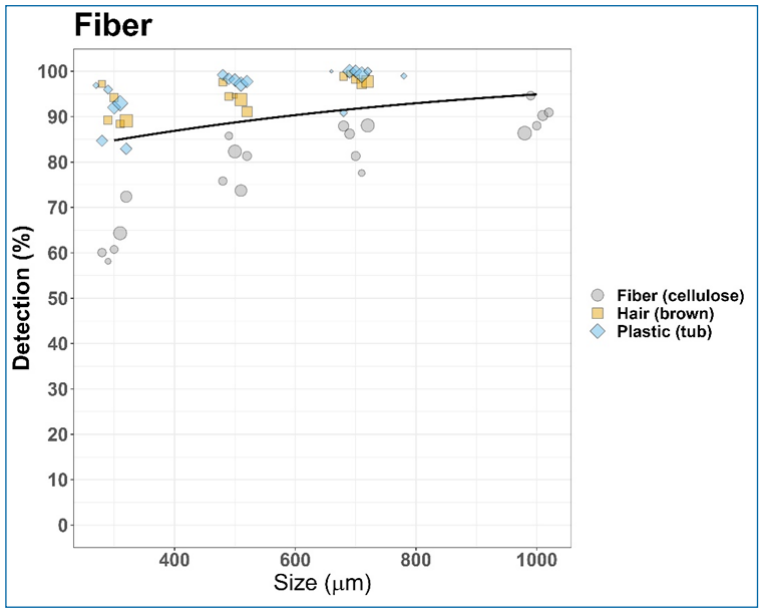

Figure C7 Probability of detection for fibers of differing type and size. Circle diameter is proportional to the number of data points falling at a specific x-y location.

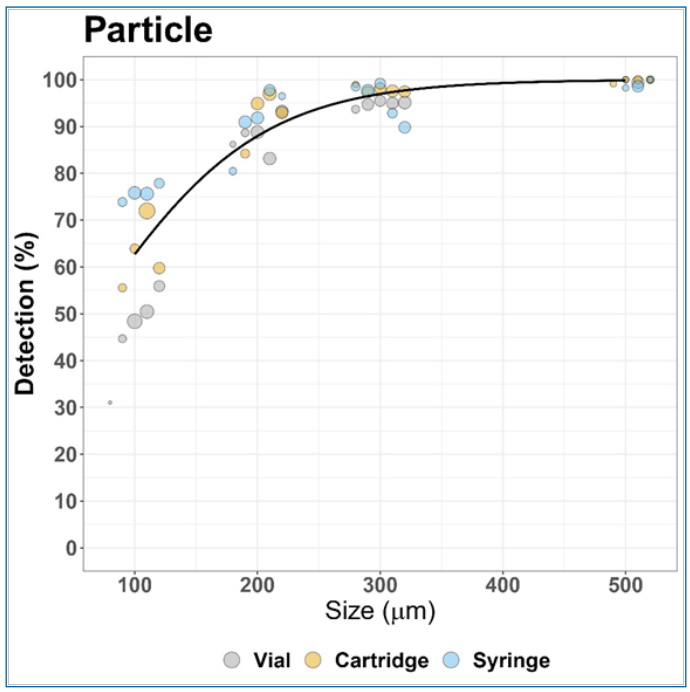

Figure C8 Probability of detection for particles in various container types and sizes. Circle diameter is proportional to the number of data points falling at a specific x-y location.

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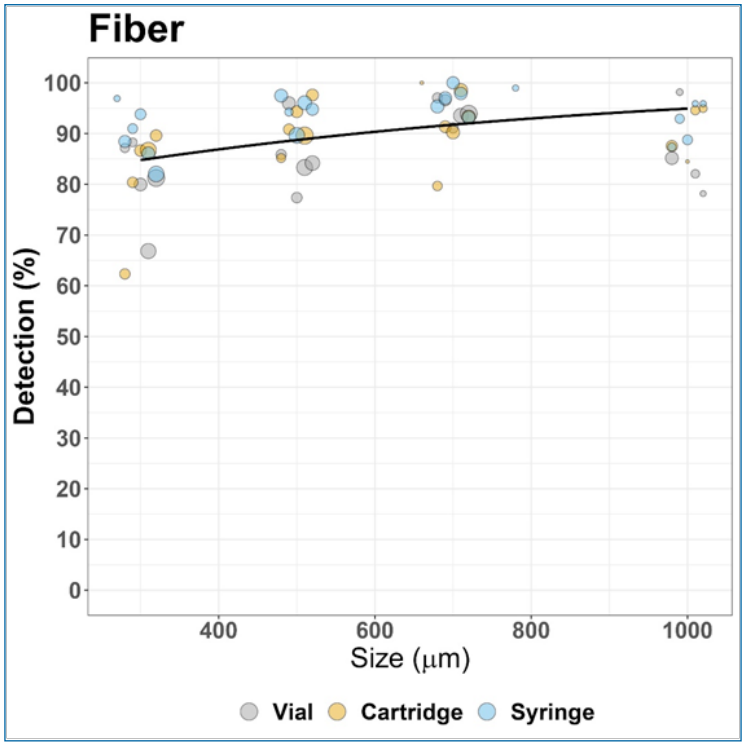

Figure C9 Probability of detection for fibers in various container types and sizes. Circle diameter is proportional to the number of data points falling at a specific x-y location.

Main Effects. Using the Boruta method *(33)* the data were analyzed for the significance that each variable has on the resulting PoD. This method is based on a machine learning technique called Random Forest. Random Forest computes the importance of predictor variables vs a response variable. It repeats this process and generates a distribution of importance for each predictor variable. It then sorts the predictor variables based on their importance and statistically compares them to random (shadow) variables to categorize them as:

- **Rejected:** variable has significantly less predictive importance than random variables.
- **Confirmed:** variable has significantly more predictive importance.
- **Tentative:** variable cannot be categorized as either rejected or confirmed when computation stops.

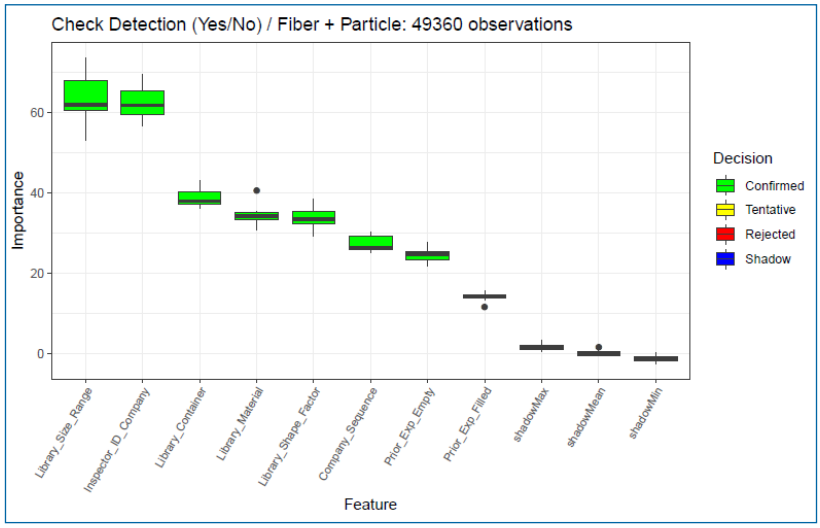

Figure C10 Boruta Plot of inspection study results demonstrating factors significant to reliable inspection performance.

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As a result, important variables are those that can play a significant role in predicting the response variable. This is similar to significant predictors in Analysis of Variance (ANOVA) but fails to satisfy usual assumptions of parametric models.

Conclusion

Based on statistical analysis of the valid data from the empty container inspection study, the main variables influencing the PoD were determined to be particle size, inspector, container, and particulate type. Further, particle size is a significant factor as well, with lower PoD observed for smaller particles. Material providing poorer contrast against the backgrounds produced lower PoD values. This was more significant when examining fibers where cellulose had a statistically lower PoD than brown hair and plastic. These results are consistent with previous visual inspection studies with filled containers.

Finally, the container size influenced the PoD, with larger containers (with a greater surface area) effectively reducing the time per unit surface area available to inspect the product and resulting in reduced detection rates for smaller particles.

8.4 Appendix D. Method: Visual Inspection of Empty Glass Containers

Introduction. The purpose of this method is to provide a procedure to identify and quantify the number of visible particles present on empty glass containers used for injectable pharmaceutical products. The procedure describes the inspection conditions and how the containers should be presented to the inspector.

Scope. This method is applicable to all empty glass containers. Inspection times may need to be increased for larger (>100mL) containers.

Equipment and materials.

1. Inspection cabinet with matte black and white backgrounds with diffuse lighting of 2,000-3,750 lux at the point of inspection.

Glass container inspection procedure.

1. Remove a container from the uninspected tray.
2. Hold the sample approximately 10 cm directly below the lamp at a distance approximately 30 cm from the inspector's eyes.
3. Inspect the sample for 5-10 sec against the black background. Rotate and invert the container to ensure all areas are examined adjusting your grip on the container as needed.
4. Inspect the sample for 5-10 sec against the white background. Rotate and invert the container to ensure all areas are examined adjusting your grip on the container as needed.
5. Place each piece examined into the either the reject tray or passing tray as required.
6. Complete any necessary records documenting the inspection of each piece before moving to the next inspection.

The time spent per container may be controlled using a pacing device. Recording the time spent inspecting each batch and then calculating the nominal inspection rate is a reliable method to confirm that the rate of inspection was maintained within the defined limits.

Information recorded during or after inspection may include details such as the type and location of any particles observed, and any other defects observed. On completion of the inspection, the data should be evaluated against the established acceptance criteria to determine the disposition of the batch. During inspection, the operator should avoid direct viewing of the light source and take regular eye breaks to avoid eye strain which may adversely influence the results of the inspection.

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